



*Charles A. Dice Center for
Research in Financial Economics*

AI Valuations: Bubble or
Fundamentals?

Kris Shen,
The Ohio State University

Dice Center WP 2026-17
Fisher College of Business WP 2026-03-017

September 2, 2026

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ABSTRACT

Do high stock valuations during the emergence of a general-purpose technology such as AI reflect price bubbles? I propose a new framework that measures the *AI premium gap*—the wedge between objective expected returns from full-information econometric benchmarks and subjective expected returns from analyst forecasts, for AI firms relative to non-AI firms. A negative gap is consistent with bubbles. From 2009 to 2024, the gap is positive. Applied to the dot-com era of the 1990s, the same framework yields a negative gap, consistent with a dot-com bubble. This contrast suggests the AI era differs fundamentally, highlighting the framework’s broader applicability.

JEL Classification: D84; G12; G14; O33

Keywords: AI; Risk premium; Subjective beliefs; Bayesian learning

*Fisher College of Business at The Ohio State University. I am deeply grateful to my dissertation committee, René Stulz (Co-Chair), Andrei Gonçalves (Co-Chair), and Aditya Chaudhry, for their continuous guidance and support. I thank Zahi Ben-David, Phil Champ, Carter Davis, Hamilton Galindo Gil, Qiang Kang (discussant), Alexander Swinton McLeod, Alessandro Melone, Sergey Sarkisyan, Daisy Wang, Michael Weisbach, Michael Wittry, Yufeng Wu, Cynthia Yin, and Shaojun Zhang for helpful discussions and comments. I am also grateful to participants at the 2026 AFA PhD Poster Session, the 2025 OSU Fisher AI in Business Conference, the 2025 SFA Annual Meeting, the 2025 Heterogeneous Agents in Asset Pricing Conference, the 2025 Finance Theory Group Conference, and the OSU Finance Student Seminar for valuable feedback.

1. Introduction

How are stocks priced when a new general-purpose technology such as AI emerges? These episodes are difficult to value in real time because their long-run cash-flow effects are highly uncertain, firms expected to benefit can experience sharp price runups, and market participants often disagree about whether prices deviate from fundamentals. AI in the early 2020s raises this question in a particularly visible way, as high stock valuations of AI firms lead to an active debate over whether the episode reflects a bubble.¹ This is not the first time a general-purpose-technology breakthrough is associated with high market valuations, which evokes déjà vu in many investors.² Despite these recurring patterns, evaluating ex ante whether stock prices during general-purpose-technology episodes reflect price bubbles remains challenging. This paper introduces a new empirical framework that helps fill this gap. I then use the framework to show that the evidence is inconsistent with the conjecture that large price increases of firms exposed to AI are caused by a bubble.

The framework compares objective expected returns, measured using econometric full-information rational-expectations benchmarks, with subjective expected returns, inferred from analyst forecasts. The logic is that bubble-like pricing is often associated with high prices supported by overly optimistic beliefs. Here, a bubble can be interpreted empirically as an episode in which asset prices rise unusually quickly and reach unusually high levels relative to fundamentals (see, e.g., [Kindleberger, 1978](#)). When prices are high relative to fundamentals, objective expected returns are low. At the same time, investors who are overly optimistic about future cash-flow growth may still view those prices as justified and perceive

¹Whether high AI stock valuations reflect a bubble is widely debated by market participants and policy institutions in the early-to-mid 2020s. On the warning side, the [Bank of England](#) argues that equity valuations, especially for AI-related technology firms, appear stretched and that correction risk has increased; the [International Monetary Fund](#) draws parallels with the 1990s dot-com bubble; and the [Bank of America Global Fund Manager Survey](#) reports that many investors view AI stocks as being in a bubble. By contrast, [Fidelity](#) argues that the AI investment cycle differs from the dot-com episode because large technology firms are largely funding AI spending with earnings rather than debt.

²Among others, [Perez \(2002\)](#) documents the recurring “passionate love affair” between financial capital and technological revolutions in five major general-purpose technologies: mechanization (1771), steam power (1829), electricity (1875), mass production (1908), and IT (1971).

expected returns to be high (see, e.g., [Greenwood and Shleifer, 2014](#); [Bordalo et al., 2024](#)). Thus, a negative gap between objective and subjective expected returns is consistent with bubble-like pricing. Following prior work, I use analyst forecasts as proxies for subjective beliefs.³

I first apply the framework to the AI era from 2009 to 2024 and then compare the results with the dot-com era in the 1990s. The choice of the AI sample period follows [Babina et al. \(2024\)](#). If high AI valuations reflect bubble-like overpricing driven by overly optimistic beliefs, subjective expected returns should exceed objective expected returns. *I find the opposite.* To fix ideas, I define the *AI premium* as the difference in expected returns between AI firms and non-AI firms. Because objective and subjective expected returns are both inferred from the same observed prices, the AI premium can be computed in two ways, one under objective expectations and the other under subjective expectations. I refer to the difference between the objective AI premium and the subjective AI premium as the *AI premium gap*. The AI premium gap is robustly positive in the data. A positive gap means that the AI premium is lower under subjective expectations than under objective expectations. Analysts perceive a smaller return advantage for AI firms relative to non-AI firms than the objective benchmark implies. By contrast, when I apply the same framework to the 1990s dot-com run-up, I find a negative *IT premium gap* between IT and non-IT firms, consistent with an ex post dot-com bubble interpretation. The contrast suggests that the AI era and the dot-com era are fundamentally different.

Because the analysis hinges on the contrast between objective and subjective expectations, I make the distinction more explicit below. Objective expected returns are full-information rational-expectations benchmarks, the best ex ante forecasts an econometrician can form using the predictive model and the information available at the time, without knowledge of subsequent realizations. Subjective expected returns are those implied by an-

³See, e.g., [De Bondt and Thaler \(1990\)](#); [La Porta \(1996\)](#); [Brav and Lehavy \(2003\)](#); [Loh and Stulz \(2018\)](#); [Bordalo et al. \(2019\)](#); [De La O and Myers \(2021\)](#); [De La O, Han, and Myers \(2022\)](#); [Nagel and Xu \(2022\)](#); [van Binsbergen, Han, and Lopez-Lira \(2023\)](#); [Birru et al. \(2024\)](#); [Bordalo et al. \(2024\)](#); [De La O and Myers \(2024\)](#); [Chaudhry \(2025\)](#).

analyst forecasts. The assumption I maintain throughout, and later formalize and assess in a Bayesian-learning model, is that analysts learn about firms' cash flows under uncertainty. They observe noisy signals about cash flows and update their beliefs gradually.

The wedge between objective and subjective expected returns can arise for two related reasons. First, analysts may not have access to the full information set or econometric tools that define the objective benchmark. Second, even when analysts observe much of the full information set, they may not fully update or process it because of cognitive, attention, or other behavioral frictions. In the first case, subjective beliefs differ from the objective benchmark because the information sets differ; in the second case, they differ because the mapping from information to expectations differs. This assumption that analysts learn under uncertainty has bite, precisely when a new general-purpose technology like AI arrives: ex ante uncertainty about its cash-flow impact is high, signals are noisy, and learning is slow. If agents can immediately and fully revise their expectations to the objective value, either because they have the full econometric model or because they perfectly process all available information, there is no AI premium gap to explain.

What does a positive AI premium gap imply about expectations of growth? A simple accounting identity makes the link transparent and turns the return-based finding into a testable prediction about analyst forecast errors. Consider the Gordon Growth Model. Since prices and dividends are observed, the same valuation can be evaluated under objective expectations, $P = D/(\mathbb{E}[r] - \mathbb{E}[g])$, or under coherent subjective expectations, $P = D/(\hat{\mathbb{E}}[r] - \hat{\mathbb{E}}[g])$, where hats denote subjective measures throughout the paper. Equating the two gives $\mathbb{E}[r] - \hat{\mathbb{E}}[r] = \mathbb{E}[g] - \hat{\mathbb{E}}[g]$, so the gap between objective and subjective expected returns is also revealing about the gap between objective and subjective expected growth.⁴

This identity is only a heuristic, but it is useful because expected growth, especially long-run expected growth, is difficult to measure directly. I measure the AI premium gap using

⁴The Gordon Growth identity is used here only to fix intuition. The same argument can be made more generally through the Campbell-Shiller valuation identity, in which case it applies to long-term expected growth rather than to growth in any particular year.

both short- and long-term expected return measures, and find a positive gap at both horizons. The growth analysis, however, does not provide an analogous comparison across horizons. Read through this lens, a positive expected return gap suggests a positive expected growth gap. For AI firms relative to other firms, objective expected growth is higher than subjective expected growth. The prediction is direct. If subjective growth expectations for AI firms are systematically below the objective benchmark, realized earnings should, on average, exceed analyst forecasts by more for AI firms than for non-AI firms. I confirm this prediction in the data. *Forecast errors*, measured as realized earnings per share (EPS) minus analyst forecasts, are more positive for AI firms relative to non-AI firms. I refer to this difference in forecast errors between AI and non-AI firms as the *forecast error gap*. It provides direct, model-free evidence for the belief pattern implied by the positive AI premium gap.⁵

The accounting identity shows what a positive AI premium gap implies, but it does not explain why the gap arises. To study the economic mechanism, I build a Bayesian learning model in which a representative analyst learns from noisy signals about cash flows. Relative to the objective full-information benchmark, the analyst updates subjective beliefs under uncertainty. Two channels, cash-flow uncertainty and positive expected cash-flow shocks, jointly explain the positive AI premium gap. First, when a new general-purpose technology like AI arrives, the analyst faces higher cash-flow uncertainty about firms that are more exposed to AI technology. Because the analyst learns gradually from noisy signals, positive but noisy cash-flow news leaves subjective expectations of future cash flows below the objective benchmark, and this wedge is larger for AI firms because their cash flows are more uncertain. It is exactly the uncertainty surrounding a large innovation shock that makes updating slow and the resulting gap large. Second, if AI firms experience more positive cash-flow news than other firms, then the wedge between objective and subjective expected cash-flow growth is larger for them even when perceived uncertainty is the same across firms.

⁵EPS and earnings forecasts are widely used to study expected growth, see, e.g., [La Porta \(1996\)](#); [Gebhardt, Lee, and Swaminathan \(2001\)](#); [Claus and Thomas \(2001\)](#); [Easton \(2004\)](#); [Ohlson and Juettner-Nauroth \(2005\)](#); [Bae, Stulz, and Tan \(2008\)](#); [Pástor, Sinha, and Swaminathan \(2008\)](#); [Hou, van Dijk, and Zhang \(2012\)](#); [Bastianello \(2025\)](#); [Gómez-Cram and Lawrence \(2025\)](#).

This channel is related in spirit to [Nagel and Xu \(2022\)](#), where investors update their beliefs about ex ante expected returns based on ex post realized outcomes, so that learning from past realizations plays a central role in shaping subjective risk premia.

To provide quantitative evidence, I structurally estimate the model and find that both channels are present. First, estimated cash-flow uncertainty is higher for AI firms than for other firms. Second, AI firms experience more positive cash-flow news over 2009–2024, especially since the launch of ChatGPT in 2022. Together, these two forces imply that subjective expectations of cash flows sit further below the objective benchmark for AI firms than for other firms, which explains the positive AI premium gap. Importantly, the expected cash-flow shocks are latent. I recover them from the model ex post, but investors cannot observe them directly in real time. This is the nature of the learning problem and the reason the gap can persist.

The learning model applies beyond AI. It requires only that a firm faces enough cash-flow uncertainty for learning to matter, and it yields conclusions similar to those here whenever two groups of firms differ sufficiently in cash-flow uncertainty and expected cash-flow shocks. For external validity, I apply the model to two additional settings, the value premium over the past four decades and the dot-com era in the 1990s. The model explains the variation in a *value premium gap*. I show that value firms have experienced more negative cash flow shocks than growth firms in the 2010s and 2020s. This sheds light on a potential reason for the declining value premium (see, e.g., [Fama and French, 2021](#); [Gonçalves and Leonard, 2023](#)). The model fails, however, to explain the variation in a similarly defined *IT premium gap* in the dot-com era. This failure is itself informative. It indicates that cash flow news, if it played any role, was not the main driver of variation in the IT premium gap during the 1990s, consistent with the ex post bubble interpretation of that era. These results show that the model is not designed for the AI setting alone. Where the model works, it identifies learning about cash flows as the main driver; where it fails, the failure lines up with the evidence of bubble-like pricing.

The model-implied evidence in my paper also relates to models of extrapolative and diagnostic beliefs (see, e.g., [Greenwood and Shleifer, 2014](#); [Barberis et al., 2018](#); [Bordalo et al., 2019](#); [Bordalo et al., 2024](#)). In these models, investors extrapolate past outcomes or overreact to news, causing subjective expectations to become too optimistic during booms and prices to rise beyond what fundamentals justify. [Greenwood and Shleifer \(2014\)](#) shows that survey return expectations are high precisely when objective expected returns are low, producing the negative gap that my framework identifies as bubble-like. The negative IT premium gap in the 1990s is consistent with this pattern, as is the failure of the cash-flow Bayesian learning model to match the empirical evidence from the dot-com era. The Bayesian model generates variation in the objective-minus-subjective spread only through learning about cash flow news. If extrapolative demand instead raised dot-com prices through subjective discount rates, as in [Barberis et al. \(2018\)](#), the IT premium gap need not comove with past cash flow news, which is what I find.

The AI evidence points in the opposite direction. A positive AI premium gap indicates that analyst forecasts of growth for AI firms fall below the objective benchmark rather than above it, and the forecast error gap provides direct evidence of this. Relative to that benchmark, the pattern is apparent underreaction rather than the overreaction associated with extrapolative bubbles ([Bordalo et al., 2019](#)). Measured against the signals analysts actually observe, the pattern requires no departure from Bayes' rule, and is what gradual updating under high cash-flow uncertainty delivers. Thus, the extrapolative belief pattern that this literature associates with bubbles may be present during the dot-com era but, so far, is absent during the AI era.

Taken together, this paper studies how stocks are priced when a new general-purpose technology arrives and ex ante uncertainty is high. A negative gap between objective and subjective expected returns is consistent with a bubble-like pricing pattern. By documenting a positive AI premium gap, I show that, if anything, the objective expected return is higher than the subjective expected return for AI firms relative to non-AI firms. This is difficult

to reconcile with a standard AI bubble interpretation. The contrast with the dot-com era sharpens the point. Even if one characterizes the boom in the dot-com era as a bubble ex post, the same pricing pattern does not appear in the AI era. I stress again that the existence of a wedge between objective and subjective expected returns does not by itself establish a bubble. A negative gap is necessary for the belief-driven bubble that this framework targets, but it is not sufficient. Prices can depart from fundamentals for reasons that have nothing to do with optimism about growth, and a negative gap can arise without a bubble. What the framework delivers is a transparent, sign-based diagnostic for whether the pricing pattern is consistent with a bubble in a given episode.

The contributions of this paper are threefold. First, this paper proposes an empirical framework for evaluating whether the pricing of general-purpose-technology episodes is consistent with a bubble story, and applies it to the AI and dot-com episodes. The practical stakes are high. When a new technology arrives, high valuations can reflect either rational repricing under uncertainty or overly optimistic beliefs. Yet the literature lacks a systematic and replicable way to bring evidence to this distinction in real time. The framework proposed here compares the gap between objective and subjective expected returns. A negative gap, where subjective expected returns exceed objective expected returns, provides evidence consistent with bubble-like pricing; a positive gap does not. This framework is not specific to AI or dot-com. It can be applied to other past and future technology episodes in which high valuations raise the question of whether prices are consistent with a bubble story.

Within the AI and asset pricing literature, the paper is the first to document a positive AI premium gap.⁶ Existing work examines how AI exposure is reflected in prices. For example, [Eisfeldt et al. \(2023\)](#) studies how generative AI affects firm-level equity returns, [Enache et al. \(2026\)](#) focuses on how the market interprets AI-related disclosures, [Ahmadi](#)

⁶There is a large literature in labor economics, macro finance, and corporate finance that studies AI-related topics, see, for example, [Cockburn, Henderson, and Stern \(2018\)](#); [Rock \(2019\)](#); [Acemoglu et al. \(2022\)](#); [Fuster et al. \(2022\)](#); [Cao et al. \(2023\)](#); [Babina et al. \(2024\)](#); [Cao et al. \(2024\)](#); [Andrews and Farboodi \(2025\)](#); [Dou, Goldstein, and Ji \(2025\)](#); [Sheng et al. \(2025\)](#); [Kung, Lustig, and Paron \(2026\)](#); [Van Nieuwerburgh \(2026\)](#); [Wachter and Wachter \(2026\)](#).

et al. (2023) studies AI-related patents, and [Gandhi et al. \(2026\)](#) shows how AI exposure affects credit spreads. This paper differs by directly measuring the wedge between objective and subjective expected returns for AI-exposed and non-AI firms, and by using this wedge to assess whether the evidence is consistent with a bubble story over the sample period.

Second, this paper contributes to the literature on subjective beliefs. A growing body of work shows that investors and analysts do not form expectations in the way fully rational models would predict (see, e.g., [De Bondt and Thaler, 1990](#); [La Porta, 1996](#); [Pástor and Veronesi, 2006](#); [Loh and Stulz, 2018](#); [Bordalo et al., 2019](#); [De La O and Myers, 2021](#); [Nagel and Xu, 2022](#); [Adam and Nagel, 2023](#); [Ben-David et al., 2023](#); [Birru et al., 2024](#); [Bordalo et al., 2024](#); [Couts, Gonçalves, and Loudis, 2024](#); [Dahlquist and Ibert, 2024](#); [De La O and Myers, 2024](#); [Décaire and Graham, 2024](#); [Jensen, 2024](#); [Bianchi et al., 2025](#); [Chaudhry, 2025](#); [Thesmar and Verner, 2025](#); [Gormsen and Huber, 2026](#)). I build on this literature by studying how belief wedges arise when investors learn about cash flows under uncertainty. The role of cash-flow uncertainty is widely discussed. For example, [Pástor and Veronesi \(2006\)](#) shows that uncertainty about future profitability can help explain high valuations of Nasdaq stocks in the late 1990s. I add a second channel based on expected cash-flow shocks, in the spirit of [Nagel and Xu \(2022\)](#), where learning from past outcomes over time shapes investors' beliefs about the future.

I combine the two channels and implement them using a simulation approach from the econometrics literature ([Carter and Kohn, 1994](#)), which is new in asset pricing. By doing so, I show that cash-flow uncertainty and expected cash-flow shocks jointly explain the variation in the AI premium gap observed in the data, and that the same mechanism helps explain asset pricing patterns in the value premium setting. In the dot-com era, by contrast, the mechanism fails to explain the IT premium gap, and the failure reinforces the bubble interpretation of that episode.

Lastly, this paper contributes to the extensive bubble literature (see, e.g., [Tirole, 1982](#); [French and Poterba, 1991](#); [Santos and Woodford, 1997](#); [Abreu and Brunnermeier, 2003](#); [Ofek](#)

and Richardson, 2003; Scheinkman and Xiong, 2003; Brunnermeier and Nagel, 2004; Pástor and Veronesi, 2006; Griffin et al., 2011; Giglio, Maggiori, and Stroebe, 2016; Barberis et al., 2018; Greenwood, Shleifer, and You, 2019; Haddad, Ho, and Loualiche, 2022). A contemporaneous paper by He et al. (2026) shows that hedge funds tilt into AI-exposed stocks. This evidence is consistent with Brunnermeier and Nagel (2004), who document that hedge funds tilted into technology stocks throughout the run-up of the dot-com bubble and only reduced their holdings shortly before the crash. However, a positive tilt with no exit is equally consistent with hedge funds currently riding an unresolved AI bubble as with no bubble at all (analogous to the discussion in Brunnermeier and Nagel, 2004).

By contrast, I contribute by providing direct evidence that contradicts an AI bubble. I directly measure the wedge between objective and subjective expected returns at the firm and aggregate levels. A positive AI premium gap indicates that objective expected growth is higher than subjective expected growth for AI firms relative to non-AI firms, which is difficult to reconcile with a simple bubble story. Applying the same framework to the dot-com era, I find a negative *IT producer premium gap* between IT producers and non-IT firms, and an insignificant *IT adopter premium gap* among IT adopters using industry IT investment. These results suggest that the AI era is fundamentally different from the dot-com era, and that while the dot-com boom can be characterized as bubble-like ex post, such a characterization does not naturally extend to AI at the time of this writing.

The rest of the paper is organized as follows. Section 2 describes the data and documents the empirical AI premium gap. Section 3 presents the analyst forecast error gap. In Section 4, I develop the Bayesian learning model. Section 5 discusses the model’s implications in the value premium and dot-com period and for the bubble debate. Section 6 concludes.

2. AI Premium Gap

As of 2026, stock market valuations of AI firms have been high. A natural reading of these valuations is a bubble one. Investors may be overly optimistic about firms that are more exposed to AI technology and bid their prices above fundamental value. For simplicity, I refer to firms with high AI exposure as AI firms. This reading is motivated by a large body of work showing that high valuations during salient growth episodes are often supported by extrapolative or overly optimistic beliefs (see, e.g., [De Bondt and Thaler, 1990](#); [La Porta, 1996](#); [Greenwood and Shleifer, 2014](#); [Bordalo et al., 2019](#); [Bordalo et al., 2024](#)).

To bring evidence to this reading, I compare two measures of expected returns for the same firm. The first is the objective expected return, the full-information rational-expectations benchmark that an econometrician forms from a predictive model and the information available at the time.⁷ The second is the subjective expected return, inferred from analyst forecasts. Following the subjective belief literature, I treat the objective measure as the benchmark and the analyst-implied measure as the belief object of interest (see, e.g., [Nagel and Xu, 2022](#)).

The economic intuition for why this comparison is informative about bubbles runs through price. Both expected returns are anchored to the same observed price, but they read that price differently. When a price is high relative to fundamentals, a larger payment buys the same future payoffs, so the objective expected return is low. An econometrician who prices the firm off fundamentals sees little compensation left in the price. An optimistic investor, however, views that same high price as justified, because she expects the firm to do better than fundamentals currently warrant. Holding the price fixed, this optimism does not lower the return she perceives; if anything she sees the high price as a fair entry point into a firm she expects to outperform, so her subjective expected return is high (see, e.g., [Greenwood and Shleifer, 2014](#); [Bordalo et al., 2024](#)). The same high price therefore pulls the objective

⁷I use the term full-information rational expectations in a looser sense than the literature does. Strictly, it requires knowing the parameters of the data-generating process. This holds for the objective benchmark in the model, but not in the empirical analysis, where the parameters must be estimated.

expected return down and leaves the subjective expected return up. The two diverge, and the direction of that divergence is the signature of bubble-like pricing. Under overoptimism, the subjective expected return exceeds the objective expected return.

To formalize the idea, I define *AI premium* as the difference in expected returns between AI firms and non-AI firms, indexed by A and N . Because objective and subjective expected returns are both inferred from the same observed prices, the AI premium can be computed in two ways. One is under objective expectations, $\mathbb{E}[r_A] - \mathbb{E}[r_N]$, and the other under subjective expectations, $\widehat{\mathbb{E}}[r_A] - \widehat{\mathbb{E}}[r_N]$, where hats denote subjective measures throughout the paper. I refer to the difference between the two as the *AI premium gap*. By the intuition above, bubble-like overpricing driven by optimism implies a negative AI premium gap, since optimism about AI firms raises the subjective AI premium relative to the objective one. The prediction need not rest on overoptimism specifically. It follows from any behavioral channel that leads investors to perceive expected returns on AI firms as unduly high given their prices.

In this section, I document the opposite empirical result. The AI premium gap is positive in the data, meaning that objective expected returns are higher than subjective expected returns for AI firms relative to non-AI firms. Through the Gordon Growth Model identity, this implies a positive wedge between objective and subjective expected growth. This is difficult to reconcile with a simple AI bubble story driven by investors' overoptimism, which would predict that subjective expected growth is too high and therefore a negative AI premium gap.

In Section 2.1, I describe the data and a firm-level *AI Exposure* measure. I use the firm-level AI Exposure to estimate the AI premium gap. I first construct the firm-level AI Exposure measure by studying how frequently AI related terms are discussed in firms' quarterly conference call transcripts. This measure is to capture how much attention investors and managers have devoted to AI-related topics. The literature has shown that using conference calls is helpful in capturing broader channels or perspectives of investors' perception

of firm risk (see, e.g., [Hassan et al., 2019](#); [Sautner et al., 2023](#)). [Eisfeldt et al. \(2023\)](#) also validates their labor-based AI Exposure measure using conference call transcripts, whereas in this paper, I directly construct AI Exposure from conference call transcripts and study the pricing implications.

In Section 2.2, I use the AI Exposure measure to estimate the AI premium gap. The estimation methodology is in the spirit of [Eskildsen et al. \(2026\)](#), but with a modified setup. While they focus on expected return measures directly, I focus on the difference between objective and subjective expected return measures. Overall, I find a positive and statistically significant AI premium gap across different expected return proxies.

2.1. AI Exposure Measure

This section describes the construction of the firm-level AI Exposure measure from 2009 to 2024.⁸ The key input is an *Artificial Intelligence Vocabulary (AIV)*, which I construct using a corpus of Web of Science academic article titles related to AI. Following the literature (see, e.g., [Hassan et al., 2019](#); [Engle et al., 2020](#); [Eisfeldt et al., 2023](#)), I extract unigrams and bigrams associated with AI topics and compute their TF-IDF scores, which measure their importance by combining how frequently they appear with how unique they are across titles. I exclude ambiguous terms and retain the most representative keywords with the highest TF-IDF scores. The detailed construction of the AIV, including the full list of terms and robustness checks using alternative corpora, is provided in the [Internet Appendix IA1](#).

Given the AIV, I construct a firm-level AI Exposure measure using firms' quarterly conference call transcripts.⁹ [Sautner et al. \(2023\)](#) uses a similar approach to measure firm-level

⁸The choice of AI sample period follows [Babina et al. \(2024\)](#). Because the data available for this paper end in 2024, the empirical results speak to the pricing of AI-exposed firms only through this period. The framework itself is the core of this paper and can be applied to later periods as more data become available.

⁹An alternative way to construct firm-level exposure is to use LLMs to assign exposure scores directly. I use a dictionary-based word-counting approach for two reasons. First, it ensures that the measure is transparent and replicable and avoids potential sensitivity to the choice of LLM used to measure AI exposure. Recent evidence suggests that LLM-based AI exposure scores can vary across models; see [Yin, Vu, and Persico \(2026\)](#). Second, it limits look-ahead bias because LLMs became available only after much of the sample period and may incorporate information from later periods.

climate change exposure. The idea in constructing this firm-level AI Exposure measure is to capture how much attention managers and investors devote to AI-related topics. Conference calls provide a natural setting where managers and investors discuss firm prospects, and prior work shows that such data capture broader perspectives on investor beliefs.

The overall firm-level AI Exposure measure is defined as the fraction of unigrams and bigrams from the AIV over all unigrams and bigrams in a transcript,

$$AIExposure_{i,t} = \frac{1}{U_{i,t} + B_{i,t}} \left(\sum_u^{U_{i,t}} \mathbb{1}[u \in AIV] + \sum_b^{B_{i,t}} \mathbb{1}[b \in AIV] \right), \quad (1)$$

where $u \in U_{i,t}$ and $b \in B_{i,t}$ denote unigrams and bigrams in the call transcript of firm i in quarter t , and $\mathbb{1}[\cdot]$ is an indicator function. The annual measure is defined as the maximum across quarters.¹⁰

In [Internet Appendix IA1](#), I also document how the *AI Exposure* measure comoves with firm-level characteristics and present examples of firms and their *AI Exposure* over the sample period. In addition, I construct related measures based on specific topics, including opportunity, regulation, and risk, as well as tone-specific measures capturing positive and negative discussions. I then compare these measures with the labor-based AI Exposure measure, $AIExposure^{Babina}$, from [Babina et al. \(2024\)](#).¹¹

Overall, the firm-level AI Exposure measure captures how much attention investors and managers devote to general AI-related topics. This measure comoves with firm characteristics such as size, BE/ME, ROE, and R&D, highlighting the importance of controlling for these firm characteristics when estimating the AI premium gap. In addition, this conference call-based AI Exposure measure comoves positively with AI labor investment as in [Babina et al. \(2024\)](#) at the aggregate level, and the AI-related discussions tend to be more positive and more focused on opportunities. In addition, this measure captures intuitive patterns, with

¹⁰Here, I do not distinguish whether AI exposure is good or bad for the firm, but rather focus on the idea that, either way, higher AI exposure is associated with greater cash-flow uncertainty surrounding AI technology advances, as explained later.

¹¹ $AIExposure^{Babina}$ from [Babina et al. \(2024\)](#) is based on AI-related labor investment up to 2018.

higher exposure among technology industries and an increasing trend over time, while also reflecting AI adoption in more traditional industries.

2.2. AI Premium Gap Estimation

To estimate the AI premium gap using the AI Exposure measure, I apply a similar panel regression used in [Eskildsen et al. \(2026\)](#). The main difference is that their dependent variable is an individual expected-return measure, whereas mine is the difference between objective and subjective expected returns, which I call the *spread*. In my setup, a panel regression is conducted to estimate the AI premium gap as the slope coefficient of all stocks' spread, denoted as $\mathbb{E}_t[r_{i,t+1}] - \widehat{\mathbb{E}}_t[r_{i,t+1}]$, on its lagged standardized AI Exposure measure.¹² The key coefficient of interest on the right-hand side is on the standardized AI Exposure measure, controlling for firm characteristics and including different sets of fixed effects, with standard errors clustered at the firm, time, and industry levels.¹³

The estimation directly tests whether there is a significant difference in the spread between low and high AI Exposure firms. I include two versions of the spread. The first version is the long-term spread which is the difference between econometric-based objective ICC as in [Gonçalves \(2021\)](#) and analyst-based subjective ICC as in [Eskildsen et al. \(2026\)](#).¹⁴ For simplicity, I refer to them as objective ICC and subjective ICC, respectively. The second version is the short-term spread which is the difference between a machine learning based expected returns using fundamentals as in [Bastianello \(2022\)](#) and analyst one-year-ahead price-target-implied expected returns as in [Loudis \(2024\)](#).¹⁵ [Internet Appendix IA3](#) docu-

¹²This approach to estimating risk premia is not new; see, for example, the carbon risk premium literature ([Bolton and Kacperczyk, 2021](#); [Bolton and Kacperczyk, 2023](#); [Zhang, 2025](#)).

¹³The detailed definition of all firm characteristics can be found in [Internet Appendix IA2](#).

¹⁴The econometric-based objective ICC is constructed by forecasting future firm fundamentals using a VAR and solving for the discount rate that equates current prices to the present value of future payouts. The analyst-based subjective ICC is the average of four different implied cost of capital measures in the accounting literature, see, [Gebhardt, Lee, and Swaminathan \(2001\)](#), [Claus and Thomas \(2001\)](#), [Ohlson and Juettner-Nauroth \(2005\)](#), and [Easton \(2004\)](#).

¹⁵The machine learning based one-year-ahead expected returns are estimated using random forest to map firm fundamentals into expected returns. The analyst one-year-ahead price-target-implied expected returns are estimated using one-year-ahead analyst price target and current price.

ments the distributions of these expected-return measures and provides details on how each is constructed.

In a panel regression setup, I regress the spread, using either its long-term or short-term version, on lagged standardized firm-level AI Exposure, standardized firm characteristics, and different sets of fixed effects (FEs):

$$Spread_{i,t} \equiv \mathbb{E}_t[r_{i,t+1}] - \widehat{\mathbb{E}}_t[r_{i,t+1}] = \text{FEs} + \beta \times AIE_{i,t-1} + \text{Controls} + \varepsilon_{i,t}, \quad (2)$$

where $AIE_{i,t-1}$ denotes lagged firm-level standardized AI Exposure, and β , which I refer to as the AI premium gap, captures the relation between lagged AI Exposure and the spread. Controls include size, BE/ME, ROE, R&D, Age, and Asset Growth. The quantitative results remain largely unchanged when additional controls are included.

AI Exposure is measured at the quarterly frequency and forward-filled to obtain a monthly series to avoid look-ahead bias. To facilitate interpretation and comparability across time, AI Exposure is standardized to have zero mean and unit standard deviation within each month. Standard errors are clustered by firm, by industry (Fama–French 49), and by month.

For robustness, I estimate three progressively more demanding specifications. The first includes month fixed effects only, the second adds firm-level controls, and the third further adds industry fixed effects. Because AI Exposure is concentrated in technologically intensive industries, the first two specifications may pick up differences across industries rather than differences in AI Exposure per se. Industry fixed effects absorb between-industry variation, so that the third specification identifies the coefficient from within-industry differences in AI Exposure and therefore provides a stricter test of the AI premium gap.

The advantage of this panel regression approach is that it allows for a richer set of control specifications, and the month fixed effects ensure that the estimated premium is not driven by aggregate shocks. At the same time, clustering standard errors at the industry level accounts

for correlated shocks across firms within the same industry, leading to more conservative inference, especially in settings where exposures differ substantially across industries, such as high-tech versus low-tech or brown versus green industries.

Figure 1 presents the estimated annual AI premium gap from the panel regressions described above, using raw spreads. Because different expected return measures have different distributions, I also report a second version of the left-hand-side spread, the spread of z-scores, which controls for differences in scale and dispersion. Each z-score is a standardized expected return measure at the monthly level. More specifically, within each month, I subtract the cross-sectional mean of the measure and divide by its cross-sectional standard deviation. Standardizing month by month puts the objective and subjective measures on a common scale, so their difference is not driven by one measure being more dispersed than the other. Those estimates are reported in Figure IA4.6 in Internet Appendix IA4 and lead to the same conclusion.¹⁶

Panel (a) shows the AI premium gap estimated using the long-term spread. The estimated AI premium gap is positive and significant across all three regression specifications. Panel (b) uses the short-term spread. With only time fixed effects, the estimated AI premium gap is insignificant. This is not surprising, since short-term expected return measures are much noisier than long-term expected return measures. That said, it also does not weaken the conclusion. A bubble requires a negative gap, so a gap that is zero is already inconsistent with a bubble. After adding controls and industry fixed effects, the AI premium gap becomes significantly positive. The same pattern holds for the spread of z-scores in Figure IA4.6.

In the specification with controls and industry fixed effects, a one-standard-deviation increase in AI Exposure is associated with an increase of about 19 basis points in the annual long-term spread and about 140 basis points in the annual short-term spread. The larger basis-point magnitude for the short-term spread mainly reflects the much higher volatility of short-term expected return measures. The z-score specifications reported in Figure IA4.6

¹⁶In Internet Appendix IA4, I also report the exact regression results for Figure 1 in Table IA4.6.

deliver the same conclusion. After adding industry fixed effects, the estimated coefficients are consistently around 0.05, so a one-standard-deviation increase in AI Exposure is associated with a 0.05-unit increase in the spread of monthly z-scores, which corresponds to about 20 basis points for the long-term spread and 187 basis points for the short-term spread.

The difference between the long-term and short-term magnitudes in basis points mainly reflects scale. The two short-term expected-return measures are much more dispersed than the two long-term measures. In [Table IA3.5](#), the objective and subjective ICCs have standard deviations of 1.66 and 4.36 percent, respectively. By comparison, the machine-learning and analyst-based short-term measures have standard deviations of 31.64 and 32.52 percent. Therefore, an effect of similar magnitude in standard-deviation terms appears as a much larger number of basis points for the short-term measures.

Are these magnitudes economically meaningful? One way to answer is to compare them with premia estimated for other firm characteristics. One recent example is the greenium. [Eskildsen et al. \(2026\)](#) estimates an annual global equity greenium of -39 basis points per standard deviation increase in standardized greenness. The long-term spread estimate for the AI premium gap is of the same order of magnitude. The AI premium gap is therefore comparable in size to a premium that the asset pricing literature treats as economically significant.

In summary, I find a positive AI premium gap using both long-term and short-term expected return measures. For complementary evidence, I include three other sets of robustness checks in [Internet Appendix IA4](#). As a first robustness check, I report the AI premium gap in [Figure IA4.7](#) estimated using expected return measures winsorized at the 99th percentile within month, to make sure the results are not driven by extreme values. The results remain largely the same. Second, I report the AI premium gap using each individual-model ICC in [Figure IA4.8](#) and [Figure IA4.9](#). The positive AI premium gap also remains largely unchanged.¹⁷ Third, I estimate the AI premium gap after adding a control variable, *isAIPro-*

¹⁷Some evidence in the literature is partially consistent with my findings. For example, [Enache et al. \(2026\)](#) uses 10-K disclosures and show that higher AI-exposure non-tech firms are associated with lower

ducer, a firm-level dummy that identifies whether a firm produces AI-related products. This check helps ensure that the results are not driven only by a few well-known AI producers like Nvidia, but instead reflect a broader cross-sectional pattern among firms with different levels of AI Exposure. This classification of AI producers is constructed using a large language model to read conference call transcripts. I identify 79 AI producer firms used in the panel regressions.¹⁸ [Figure IA4.10](#) reports the estimation results that are largely the same.

3. Analyst Forecast Error Gap

In the previous section, I document a positive *AI premium gap*, meaning that objective expected returns are higher than subjective expected returns for high *AI Exposure* firms relative to low *AI Exposure* firms. Through the Gordon Growth Model identity, this return wedge should be reflected in a corresponding wedge in expected growth. Formally, holding the observed price and dividend fixed, the identity implies

$$\mathbb{E}[r] - \widehat{\mathbb{E}}[r] = \mathbb{E}[g] - \widehat{\mathbb{E}}[g]. \quad (3)$$

Therefore, a positive AI premium gap implies a positive wedge between objective and subjective expected growth, that is,

$$\mathbb{E}[g] - \widehat{\mathbb{E}}[g] > 0. \quad (4)$$

Directly measuring this expected growth wedge is difficult, because there is no generally available measure of long-run expected growth. The measure normally used is the Institutional Brokers' Estimate System (IBES) long-term growth forecast, *LTG*, which covers a horizon of three to five years (see, e.g., [Bordalo et al., 2024](#)). Even with LTG, each observation would require at least three years of subsequent realized growth, which is a problem given the short sample in the AI era. Instead, I provide evidence using short-run realized

analyst-based subjective ICC.

¹⁸The exact LLM prompt used to classify AI producer firms is documented in [Internet Appendix IA6](#).

forecast errors from analyst earnings forecasts in IBES databases. If subjective growth expectations for high AI Exposure firms are systematically below the objective benchmark, then realized earnings growth should be higher than analysts expected, especially for firms with higher AI Exposure. This gives a testable implication of the valuation identity, that high AI Exposure firms should have more positive analyst forecast errors relative to low AI Exposure firms.

3.1. *Forecast Error Gap*

In this section, I test this implication. I examine whether firms with higher AI Exposure experience more positive *forecast errors*, defined as the difference between realized earnings per share (EPS) and analyst forecasts. I find that is the case. This finding provides short-run evidence consistent with the growth-side implication of the positive AI premium gap.

More specifically, I estimate analyst forecast bias following [Gómez-Cram and Lawrence \(2025\)](#). Forecast errors are defined as the difference between realized EPS and analyst consensus forecasts at horizon h , where h runs from one to four quarters ahead:

$$e_{i,t+h|t} \equiv y_{i,t+h} - \mathbb{F}_t y_{i,t+h} = a_h + \beta_h^{AIE} AIE_{i,t} + u_{i,t+h}, \quad (5)$$

where $y_{i,t+h}$ denotes realized EPS for firm i at time $t + h$, and $\mathbb{F}_t y_{i,t+h}$ is the analyst forecast formed at time t . The parameter a_h captures the average forecast bias at horizon h . Standard errors are clustered by firm and calendar year-quarter. $AIE_{i,t}$ is the standardized AI Exposure measure prior to forecast issuance. The coefficient β_h^{AIE} captures how analyst forecast errors vary with AI Exposure.¹⁹

For each firm-quarter, I include all analyst forecasts issued within a 45-day window prior to the earnings announcement and construct the consensus forecast as the median across analysts. I exclude micro-cap firms and winsorize forecast errors at the 99.9th percentile

¹⁹I re-estimate (5) with the forecast error scaled by the lagged stock price and, separately, by book equity per share, and obtain similar results.

separately for each horizon.

[Table 1](#) reports the estimation results in cents per share. Analysts systematically underestimate the earnings of firms with higher AI Exposure. Column (1) shows that the estimated a_h is negative and significant at horizons of 2, 3, and 4 quarters, consistent with the literature that analysts tend to overestimate earnings for average firms.²⁰ For a firm with average AI Exposure, realized earnings fall short of the analyst forecast by 10.09 cents per share at the two-quarter horizon, and the shortfall widens to 21.66 cents per share at the four-quarter horizon. In contrast, Column (2) shows that the estimated β_h^{AIE} is positive and significant across all horizons, indicating that analysts underestimate earnings for firms with higher AI Exposure. A one-standard-deviation increase in AI Exposure is associated with realized earnings that exceed the analyst forecast by an additional 12.65 cents per share at the one-quarter horizon, rising to 18.23 cents per share at the four-quarter horizon. This pattern is consistent with the Gordon Growth Model identity.

For robustness checks, I estimate forecast bias separately for AI producers, non-AI producers, and their difference in [Internet Appendix IA7](#). The classification of AI producer firms is based on LLMs reading conference call transcripts to identify whether firms produce AI-related products. [Table IA7.7](#) reports the estimation results. For AI producers, forecast bias is positive and statistically significant at short horizons, indicating that realized earnings exceed analyst forecasts on average. In contrast, forecast bias for non-AI producers is negative and highly significant across all horizons, consistent with analyst overestimation. The difference between the two groups is positive and statistically significant across all horizons, with the magnitude increasing with horizon, consistent with [Table 1](#).

Overall, in this section, I show that there is a positive forecast error gap, meaning that analysts tend to underestimate earnings for AI firms, while overestimating earnings for non-AI firms. This positive forecast error gap, together with the positive AI premium gap, is consistent with the Gordon Growth Model identity as discussed in the beginning of this

²⁰Analysts tend to overestimate earnings for average firms, see, e.g., [Michaely and Womack \(1999\)](#); [Lim \(2001\)](#); [Hong and Kubik \(2003\)](#); [Malmendier and Shanthikumar \(2007\)](#); [Hou, van Dijk, and Zhang \(2012\)](#).

section. The next section provides a model with Bayesian investors to explain my empirical findings on the AI premium gap and forecast error gap.

4. Model

After documenting the *AI premium gap* and the analyst *forecast error gap*, I now build a Bayesian learning model to understand the mechanism behind these findings. The Gordon Growth Model is useful for guiding the empirical tests, but it is static and does not speak directly to the underlying mechanism. [Section 4.1](#) starts with a simple two-period model to illustrate the key mechanism, that is the positive AI premium gap arises from the interaction between cash-flow uncertainty and positive expected cash flow shocks.²¹ Uncertainty alone does not generate the gap without positive expected cash flow shocks, and positive expected cash flow shocks do not generate the gap without uncertainty. It is the interaction of the two that creates a wedge between the econometric-based objective and analyst-based subjective expected returns.

[Section 4.2](#) then extends the two-period model to a dynamic setting and shows quantitative results consistent with this mechanism. The core output of the model is the AI premium gap between objective expected returns under full-information rational expectations, or FIRE, and subjective expected returns formed by analysts through Bayesian learning. This corresponds in the empirical analysis to the wedge between econometric-based objective expected returns and analyst-based subjective expected returns.²² In [Section 1](#), I have

²¹To be more precise, positive expected cash flow shocks refer to the difference between the objective and subjective expectations of latent expected cash flows in the two-period model, and to the difference between objective and subjective expected latent dividend growth in the dynamic model.

²²In the empirical analysis, econometric-based objective expected returns are constructed using past data. In principle, analysts could form similar expectations using all publicly available information. The model should therefore be interpreted as capturing a setting in which analysts observe noisy signals and may not fully process all information available from firms, possibly due to cognitive or attention constraints. Bayesian updating provides a tractable benchmark for this belief formation process. The key ingredient, however, is uncertainty rather than the exact Bayesian-optimal updating rule. The mechanism continues to apply as long as beliefs adjust gradually, even if the degree of updating deviates from the Bayesian-optimal Kalman gain. [Chaudhry \(2025\)](#) demonstrates this point formally by introducing behavioral biases that generate departures from the Kalman gain, and shows that the core theoretical results are preserved under such deviations.

discussed two reasons why objective and subjective expected returns can differ. Analysts may not have the full information set, or they may not fully process the information they do have. The model in this section takes the first route and lets a single friction, cash-flow uncertainty, do all the work. I make this choice for parsimony. It keeps the mechanism transparent and the parameter space small enough to estimate. It is not a claim that the second route is absent. If analysts update more slowly than Bayes' rule prescribes, subjective expectations sit further from the objective benchmark, so cognitive and attention frictions widen the wedge rather than reverse it. Detailed model derivations can be found in [Internet Appendix IA8](#).

4.1. *Two-Period Illustrative Model*

This section presents a two-period Bayesian learning model that illustrates the mechanism in its simplest form. The model distinguishes the objective expected return under full information from the investor's subjective expected return when the expected cash flow component is not directly observed. Their difference is linked to the forecast error gap.

4.1.1. **Environment and Bayesian learning**

There is one representative investor that is a Bayesian learner. There are two dates, $t = 0, 1$, and two risky assets indexed by $i \in \{A, N\}$, where A denotes an AI firm and N denotes a non-AI firm. At date $t = 1$, asset i pays a terminal dividend

$$D_{i,1} = \bar{D}_i + \eta_{i,0} + \varepsilon_{i,1}, \quad (6)$$

where $\eta_{i,0}$ is an unobserved cash flow component and $\varepsilon_{i,1}$ is a firm-specific dividend shock. I assume

$$\eta_{i,0} \sim \mathcal{N}(0, \sigma_{\eta_i}^2), \quad \varepsilon_{i,1} \sim \mathcal{N}(0, \sigma_{\varepsilon i}^2). \quad (7)$$

At date $t = 0$, the investor observes a noisy signal

$$s_{i,0} = \eta_{i,0} + u_{i,0}, \quad u_{i,0} \sim \mathcal{N}(0, \sigma_{ui}^2), \quad (8)$$

where $\eta_{i,0}$, $\varepsilon_{i,1}$, and $u_{i,0}$ are mutually independent for each asset and across assets. I denote the investor's information set by $\widehat{\mathcal{I}}_{i,0}$. It contains the signal and the known model parameters but excludes $\eta_{i,0}$. That means, the investor does not observe $\eta_{i,0}$ but knows the true prior distribution for it as in (7). The full information set $\mathcal{I}_{i,0}$ additionally includes $\eta_{i,0}$. Gaussian updating gives

$$\widehat{\mathbb{E}}[\eta_{i,0} \mid \widehat{\mathcal{I}}_{i,0}] = G_{i,0}s_{i,0}, \quad (9)$$

where

$$G_{i,0} \equiv \frac{\sigma_{\eta i}^2}{\sigma_{\eta i}^2 + \sigma_{ui}^2} \quad (10)$$

is the Kalman gain. The posterior variance is

$$\widehat{\mathbb{V}}_{i,0} \equiv \widehat{\mathbb{V}}(\eta_{i,0} \mid \widehat{\mathcal{I}}_{i,0}) = (1 - G_{i,0})\sigma_{\eta i}^2, \quad (11)$$

and the conditional variance of the terminal dividend is

$$\widehat{\Omega}_{i,0} \equiv \widehat{\mathbb{V}}(D_{i,1} \mid \widehat{\mathcal{I}}_{i,0}) = \widehat{\mathbb{V}}_{i,0} + \sigma_{\varepsilon i}^2. \quad (12)$$

4.1.2. Equilibrium expected returns and gaps

The investor has mean-variance preferences with risk aversion γ and there is one unit of each asset in net supply. Market clearing gives the equilibrium price

$$P_{i,0} = \bar{D}_i + G_{i,0}s_{i,0} - \gamma\widehat{\Omega}_{i,0}. \quad (13)$$

The subjective and objective expected dollar returns are therefore

$$\widehat{\mathbb{E}}[D_{i,1} - P_{i,0} \mid \widehat{\mathcal{I}}_{i,0}] = \gamma \widehat{\Omega}_{i,0}, \quad (14)$$

$$\mathbb{E}[D_{i,1} - P_{i,0} \mid \mathcal{I}_{i,0}] = \gamma \widehat{\Omega}_{i,0} + \eta_{i,0} - G_{i,0} s_{i,0}. \quad (15)$$

The objective-minus-subjective expected-return gap for asset i is thus the belief error $\eta_{i,0} - G_{i,0} s_{i,0}$. Differencing this gap across AI and non-AI firms gives the *AI premium gap*:

$$\begin{aligned} \text{Gap}_0 &= (\eta_{A,0} - G_{A,0} s_{A,0}) - (\eta_{N,0} - G_{N,0} s_{N,0}) \\ &= (1 - G_{A,0}) \eta_{A,0} - (1 - G_{N,0}) \eta_{N,0} - G_{A,0} u_{A,0} + G_{N,0} u_{N,0}. \end{aligned} \quad (16)$$

The risk-premium component enters both objective and subjective expected returns and therefore cancels from the gap. The forecast error for asset i is

$$FE_{i,1} = (1 - G_{i,0}) \eta_{i,0} - G_{i,0} u_{i,0} + \varepsilon_{i,1}. \quad (17)$$

It follows that

$$FE_{A,1} - FE_{N,1} = \text{Gap}_0 + \varepsilon_{A,1} - \varepsilon_{N,1}. \quad (18)$$

Thus, the AI premium gap maps into the forecast error gap up to the difference in firm-specific dividend shocks.

4.1.3. The interaction of two channels

The model highlights the interaction between Channel 1, cash-flow uncertainty, and Channel 2, positive expected cash flow shocks. Signal noise lowers the Kalman gain and makes the investor place less weight on observed cash flow news. Positive latent expected cash flows then generate a larger belief error. Neither channel alone is sufficient to generate a positive gap.

For Channel 1, suppose $\eta_{A,0} = \eta_{N,0} = \eta_0 > 0$ and the realized signal noise is negligible. Then

$$\text{Gap}_0 \approx (G_{N,0} - G_{A,0})\eta_0 > 0 \quad \text{if } G_{A,0} < G_{N,0}. \quad (19)$$

A noisier AI signal lowers $G_{A,0}$ and increases the AI premium gap, but only when the latent expected cash flow component is positive.

For Channel 2, suppose $G_{A,0} = G_{N,0} = G_0$ and the realized signal noise is negligible. Then

$$\text{Gap}_0 \approx (1 - G_0)(\eta_{A,0} - \eta_{N,0}) > 0 \quad \text{if } \eta_{A,0} > \eta_{N,0}. \quad (20)$$

More positive latent expected cash flows for AI firms increase the AI premium gap, but only when signals are noisy, so that even a Bayesian analyst updates gradually. [Figure 2](#) illustrates these two cases using values that are illustrative and not calibrated. Taken together, the positive AI premium gap arises when positive expected cash flow shocks occur in a setting where cash flow uncertainty is high and investors therefore learn gradually.

4.2. *Dynamic Model*

This section extends the two-period illustrative model to a dynamic setting with sequential Bayesian learning. I use GMM-based structural estimation to provide quantitative evidence for Channel 1 and a Carter-Kohn backward smoothing simulation following [Carter and Kohn \(1994\)](#) to assess Channel 2. Detailed derivations are provided in [Internet Appendix IA8](#).

4.2.1. **Environment and Bayesian learning**

There are two risky assets, indexed by $i \in \{A, N\}$, corresponding to AI and non-AI firms. A representative investor prices the assets while learning about a latent expected cash flow state. Following [De La O, Han, and Myers \(2022\)](#) and [Chaudhry \(2025\)](#), the log stochastic

discount factor is

$$m_{t+1} = -r_f - \frac{1}{2}\gamma^2\sigma^2 - \gamma\varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2), \quad (21)$$

where r_f is the log risk-free rate, γ is risk aversion, and ε_{t+1} is an aggregate shock. The investor's Euler equation is

$$\widehat{\mathbb{E}}_t[\exp(m_{t+1})R_{i,t+1}] = 1, \quad r_{i,t+1} \equiv \log R_{i,t+1}. \quad (22)$$

In a similar spirit to [van Binsbergen and Koijen \(2010\)](#), dividend growth and the latent state evolve as

$$\Delta d_{i,t+1} = \mu_d + \eta_{i,t} + \beta_i \varepsilon_{t+1} + u_{i,t+1}, \quad (23)$$

$$\eta_{i,t+1} = \rho_i \eta_{i,t} + \epsilon_{i,t+1}, \quad \epsilon_{i,t+1} \sim \mathcal{N}(0, \sigma_{i,\epsilon}^2). \quad (24)$$

The idiosyncratic cash flow shock satisfies

$$u_{i,t+1} \sim \mathcal{N}(0, \sigma_{i,u}^2). \quad (25)$$

I assume $|\rho_i| < 1$ and initialize the latent state at its stationary distribution. Each shock sequence is i.i.d. over time. The aggregate shock is independent of the asset-specific cash flow and state innovations, which are also mutually independent across assets. The investor observes realized dividend growth but does not separately observe the individual shocks.

For tractability, the investor learns about each asset separately using only the history of that asset's realized dividend growth. The signal observed at date $t + 1$ is

$$s_{i,t+1} \equiv \Delta d_{i,t+1} - \mu_d = \eta_{i,t} + \nu_{i,t+1}, \quad (26)$$

where

$$\nu_{i,t+1} \equiv \beta_i \varepsilon_{t+1} + u_{i,t+1}, \quad (27)$$

with variance

$$U_i \equiv \widehat{\mathbb{V}}(\nu_{i,t+1}) = \beta_i^2 \sigma^2 + \sigma_{i,u}^2. \quad (28)$$

Thus, the signal observed at $t + 1$ provides information about $\eta_{i,t}$. Let $\widehat{\mathcal{I}}_{i,t}$ denote the investor's information set generated by that asset's signal history through date t and the known model parameters. I define

$$\widehat{\eta}_{i,t} \equiv \widehat{\mathbb{E}}[\eta_{i,t} \mid \widehat{\mathcal{I}}_{i,t}], \quad P_{i,t} \equiv \widehat{\mathbb{V}}(\eta_{i,t} \mid \widehat{\mathcal{I}}_{i,t}). \quad (29)$$

Because the signal about $\eta_{i,t}$ arrives at $t + 1$, the investor first updates beliefs about $\eta_{i,t}$ and then uses the state transition to form beliefs about $\eta_{i,t+1}$. Following [Hamilton \(1994\)](#), the Kalman gain and the resulting recursions can be written down as

$$G_{i,t} = \frac{P_{i,t}}{P_{i,t} + U_i}, \quad (30)$$

$$\widehat{\eta}_{i,t+1} = \rho_i [\widehat{\eta}_{i,t} + G_{i,t} (s_{i,t+1} - \widehat{\eta}_{i,t})], \quad (31)$$

$$P_{i,t+1} = \rho_i^2 (1 - G_{i,t}) P_{i,t} + \sigma_{i,\epsilon}^2. \quad (32)$$

4.2.2. Equilibrium prices and expected returns

I use the log-linear Campbell–Shiller identity as in [Campbell and Shiller \(1988\)](#),

$$r_{i,t+1} = \kappa_0 + \kappa_1 z_{i,t+1} - z_{i,t} + \Delta d_{i,t+1}, \quad (33)$$

where $z_{i,t}$ is the log price-dividend ratio. Because the investor sets prices using $\widehat{\mathcal{I}}_{i,t}$, I conjecture that

$$z_{i,t} = A_{0,i} + A_{1,i} \widehat{\eta}_{i,t}. \quad (34)$$

The objective full-information benchmark does not enter this pricing rule. I focus on the steady state of the Kalman filter. Let P_i^* and G_i^* satisfy

$$G_i^* = \frac{P_i^*}{P_i^* + U_i}, \quad P_i^* = \rho_i^2(1 - G_i^*)P_i^* + \sigma_{i,\epsilon}^2. \quad (35)$$

The equilibrium coefficients are

$$A_{1,i} = \frac{1}{1 - \kappa_1 \rho_i}, \quad (36)$$

and

$$B_i^* \equiv 1 + \kappa_1 A_{1,i} \rho_i G_i^*. \quad (37)$$

The term B_i^* captures the direct effect of cash flow news on returns and the additional price response generated by belief updating. The intercept $A_{0,i}$ is pinned down by the Euler equation and is derived in [Internet Appendix IA8](#). The resulting return decomposition is

$$r_{i,t+1} = C_i + B_i^* (\eta_{i,t} - \hat{\eta}_{i,t}) + B_i^* (\beta_i \varepsilon_{t+1} + u_{i,t+1}), \quad (38)$$

where $C_i \equiv \kappa_0 + (\kappa_1 - 1)A_{0,i} + \mu_d$. The belief error is uncertain to the investor but known under full information. The remaining shocks are unforecastable under both information sets. In the spirit of [Nagel and Xu \(2022\)](#), the investor's subjective expected log excess return is

$$\hat{\mathbb{E}}_t[r_{i,t+1} - r_f] = C_i - r_f = \gamma B_i^* \beta_i \sigma^2 - \frac{1}{2}(B_i^*)^2(P_i^* + U_i). \quad (39)$$

Let $\mathcal{I}_{i,t}$ denote the full information set, which additionally reveals $\eta_{i,t}$. The objective expectation is a full-information econometric benchmark and does not determine prices. Conditioning the same equilibrium return on $\mathcal{I}_{i,t}$ gives

$$\mathbb{E}_t[r_{i,t+1} - r_f] = \hat{\mathbb{E}}_t[r_{i,t+1} - r_f] + B_i^* (\eta_{i,t} - \hat{\eta}_{i,t}). \quad (40)$$

Therefore, the objective-minus-subjective expected-return gap is

$$\mathbb{E}_t[r_{i,t+1} - r_f] - \widehat{\mathbb{E}}_t[r_{i,t+1} - r_f] = B_i^* (\eta_{i,t} - \widehat{\eta}_{i,t}). \quad (41)$$

This is the central mechanism of the model. The gap is positive when the true latent expected cash flow state exceeds the investor's conditional mean. Under Bayesian learning, the gap is zero on average, but it varies over time with the belief error.

4.2.3. AI premium and forecast error gaps

Define the AI premium under objective and subjective expectations as

$$\Pi_t^{obj} \equiv \mathbb{E}_t[r_{A,t+1} - r_f] - \mathbb{E}_t[r_{N,t+1} - r_f], \quad (42)$$

$$\Pi_t^{sub} \equiv \widehat{\mathbb{E}}_t[r_{A,t+1} - r_f] - \widehat{\mathbb{E}}_t[r_{N,t+1} - r_f]. \quad (43)$$

The *AI premium gap* is

$$\Pi_t^{gap} \equiv \Pi_t^{obj} - \Pi_t^{sub} = B_A^* (\eta_{A,t} - \widehat{\eta}_{A,t}) - B_N^* (\eta_{N,t} - \widehat{\eta}_{N,t}). \quad (44)$$

A positive AI premium gap arises when the return-scaled belief error is larger for AI firms than for non-AI firms. The gap is mean zero unconditionally, but it is persistent. The Kalman recursion in (31) implies that the belief error $e_{i,t} \equiv \eta_{i,t} - \widehat{\eta}_{i,t}$ is itself an AR(1),

$$e_{i,t+1} = \rho_i (1 - G_i^*) e_{i,t} + \epsilon_{i,t+1} - \rho_i G_i^* (\beta_i \varepsilon_{t+1} + u_{i,t+1}). \quad (45)$$

Because the AI premium gap in (44) is a fixed combination of the two belief errors, it inherits the persistence $\rho_i(1 - G_i^*)$. Two forces could make this coefficient large. The latent state could be persistent, and noisy signals might lower the Kalman gain, so beliefs might correct slowly. Signal noise therefore governs both how far subjective expectations sit below the

objective benchmark and how long they stay there.

The investor's one-step-ahead forecast of dividend growth is

$$\widehat{\mathbb{E}}_t[\Delta d_{i,t+1}] = \mu_d + \widehat{\eta}_{i,t}, \quad (46)$$

so the forecast error is

$$FE_{i,t+1} = (\eta_{i,t} - \widehat{\eta}_{i,t}) + \beta_i \varepsilon_{t+1} + u_{i,t+1}. \quad (47)$$

Under full information,

$$\mathbb{E}_t[FE_{i,t+1}] = \eta_{i,t} - \widehat{\eta}_{i,t}, \quad (48)$$

and therefore

$$\mathbb{E}_t[r_{i,t+1} - r_f] - \widehat{\mathbb{E}}_t[r_{i,t+1} - r_f] = B_i^* \mathbb{E}_t[FE_{i,t+1}]. \quad (49)$$

The same belief error generates both the expected-return gap and the objective expected forecast error. For the GMM estimation, I use the second moment of the forecast error since the hidden state, $\eta_{i,t}$, is not observable in the above first moment. In the steady state,

$$\widehat{\mathbb{E}}_t[FE_{i,t+1}^2] = P_i^* + U_i. \quad (50)$$

The steady-state mean squared forecast error gap is

$$\text{FEGap}^{MSE} = (P_A^* + U_A) - (P_N^* + U_N). \quad (51)$$

The distribution of the AI premium gap and the mean squared forecast error gap is governed by the same structural primitives. Higher measurement noise increases U_i , lowers the Kalman gain, and slows learning. Greater state innovation variance raises P_i^* . The realized AI premium gap additionally depends on the realized belief errors for AI and non-AI firms. The GMM estimation below disciplines Channel 1, while the Carter-Kohn simulation recovers

latent-state paths to assess Channel 2.

4.3. GMM Estimation

4.3.1. Estimation details

I estimate the structural parameters of the dynamic learning model using GMM, following the moment-based estimation approach widely used in asset pricing literature (see, e.g., [Hansen, 1982](#); [Hansen and Singleton, 1982](#); [Jagannathan, Skoulakis, and Wang, 2002](#); [Cochrane, 2005](#); [Gomes, Yaron, and Zhang, 2006](#); [Liu, Whited, and Zhang, 2009](#); [Belo, Deng, and Salomao, 2024](#); [Chaudhry, 2025](#)). The estimation uses the US data from 2009 to 2024 and distinguishes AI and non-AI firms using the highest and lowest *AI Exposure* quintiles. The model is disciplined using calibrated parameters and empirical target moments, as summarized in [Table 2](#). The details about the GMM estimation can be found in [Internet Appendix IA9](#).

I calibrate seven parameters using data from 2009 to 2024. As in [Chaudhry \(2025\)](#), the log risk-free rate is calibrated to the average one year Treasury bill return from the CRSP fixed term indices, yielding $r_f = 0.0117$. The aggregate shock volatility is calibrated to the annualized volatility of CRSP value-weighted market returns, yielding $\sigma = 0.1804$. The mean dividend growth is calibrated from four-year firm-level dividend growth, $G_{n,t} = \sum_{s=1}^4 \Delta d_{n,t+s}$, winsorized at five percent, by computing firm level annual geometric mean growth rates and taking the pooled average, yielding $\mu_d = 0.0683$. The Campbell Shiller constants are calibrated from the sample mean log price dividend ratio $\bar{z} = \mathbb{E}[\log(P/D)]$ through

$$\kappa_1 = \frac{\exp(\bar{z})}{1 + \exp(\bar{z})}, \quad \kappa_0 = -\log(1 - \kappa_1) - \kappa_1 \bar{z}, \quad (52)$$

yielding $\kappa_0 = 0.0743$ and $\kappa_1 = 0.9858$. Market betas are estimated separately for AI and non-AI firms, yielding $\beta_A = 0.9414$ and $\beta_N = 0.8835$.

The remaining parameters are estimated by matching model implied moments under the model steady state distribution. For each group $i \in \{A, N\}$, the model moments are tied

to the following objects. First, subjective expected log excess returns are matched using value-weighted analyst-based subjective ICCs for portfolios sorted by AI Exposure, with the highest quintile classified as AI firms and the lowest quintile classified as non-AI firms. The model counterpart is

$$\widehat{\mathbb{E}}_t[r_{i,t+1} - r_f] = \gamma B_i^* \beta_i \sigma^2 - \frac{1}{2} (B_i^*)^2 (U_i + P_i^*), \quad (53)$$

with empirical targets $\widehat{\mathbb{E}}[r_{A,t+1} - r_f] = 0.0682$ and $\widehat{\mathbb{E}}[r_{N,t+1} - r_f] = 0.0737$.

Second, the unconditional variance of the log price-dividend ratio (winsorized at 5%) for AI and non-AI firms is matched. For $\forall i \in \{A, N\}$, define $z_{i,t} \equiv \log(P_{i,t}/D_{i,t})$. Under Campbell–Shiller and the conjecture,

$$z_{i,t} = A_{0,i} + A_{1,i} \widehat{\eta}_{i,t}, \quad A_{1,i} = \frac{1}{1 - \kappa_1 \rho_i}. \quad (54)$$

Since $\widehat{\eta}_{i,t} = \widehat{\mathbb{E}}[\eta_{i,t} | \widehat{\mathcal{I}}_{i,t}]$ and $\mathbb{V}(\eta_{i,t}) = \sigma_{i,\epsilon}^2 / (1 - \rho_i^2)$, the law of total variance implies

$$\mathbb{V}(\widehat{\eta}_{i,t}) = \mathbb{V}(\eta_{i,t}) - \mathbb{E} \left[\widehat{\mathbb{V}}(\eta_{i,t} | \widehat{\mathcal{I}}_{i,t}) \right] = \frac{\sigma_{i,\epsilon}^2}{1 - \rho_i^2} - P_i^*, \quad (55)$$

where $P_i^* \equiv \widehat{\mathbb{V}}(\eta_{i,t} | \widehat{\mathcal{I}}_{i,t})$ is the steady-state conditional variance of the current latent state and $U_i \equiv \beta_i^2 \sigma^2 + \sigma_{i,u}^2$. Therefore, the model implies

$$\mathbb{V}(z_{i,t}) = A_{1,i}^2 \left(\frac{\sigma_{i,\epsilon}^2}{1 - \rho_i^2} - P_i^* \right). \quad (56)$$

Using data from 2009 to 2024, I obtain $\mathbb{V}(z_{A,t}) = 0.9813$ and $\mathbb{V}(z_{N,t}) = 1.0560$.

Third, the unconditional variance of the objective-minus-subjective spread is matched. Under stationarity,

$$\mathbb{V}(\text{Spread}_{i,t}) = (B_i^*)^2 P_i^*, \quad (57)$$

where P_i^* is the steady-state conditional variance from the scalar Kalman filter and B_i^* is the

steady-state return transmission coefficient. The empirical targets are $\mathbb{V}(\text{Spread}_{A,t}) = 0.0285$ and $\mathbb{V}(\text{Spread}_{N,t}) = 0.0151$.

Lastly, mean squared forecast errors are matched using

$$\widehat{\mathbb{E}}_t[FE_{i,t+1}^2] = P_i^* + U_i, \quad (58)$$

with empirical targets $\widehat{\mathbb{E}}[FE_{A,t+1}^2] = 0.1805$ and $\widehat{\mathbb{E}}[FE_{N,t+1}^2] = 0.1223$.

4.3.2. Estimation results and Channel 1 discussion

The GMM estimation yields the structural parameter estimates reported in [Table 3](#). The point estimates provide suggestive evidence consistent with Channel 1 through the signal-noise margin. The estimated idiosyncratic observation noise variance is higher for AI firms than for non-AI firms, with $\sigma_{A,u}^2 = 0.1340$ and $\sigma_{N,u}^2 = 0.1036$. Together with the higher market beta of AI firms, $\beta_A = 0.9414$ compared with $\beta_N = 0.8835$, this gives total signal noise variances of $U_A = 0.1628$ and $U_N = 0.1290$. The implied steady-state Kalman gain is correspondingly lower for AI firms, with $G_A^* = 0.0357$ compared with $G_N^* = 0.0511$. Thus, investors place less weight on newly observed cash flow news about AI firms.

The estimates also imply that the gap can be persistent. The implied AR(1) coefficient of the belief error, $\rho_i(1 - G_i^*)$, is 0.9273 for AI firms and 0.9059 for non-AI firms, with half-lives of about 9.2 and 7.0 years. The gap combines the two belief errors, so both coefficients govern its decay, and the AI root dominates at long horizons. A low Kalman gain and a persistent latent state therefore show up not only in slower updating at a point in time, but also in a longer-lived gap.

However, the other structural parameters do not all point toward greater latent-state uncertainty for AI firms. The state innovation variance is lower for AI firms, with $\sigma_{A,\epsilon}^2 = 0.0007$ compared with $\sigma_{N,\epsilon}^2 = 0.0009$. AI firms also have slightly higher state persistence, with $\rho_A = 0.9616$ compared with $\rho_N = 0.9547$, and lower steady-state conditional variance,

with $P_A^* = 0.0060$ compared with $P_N^* = 0.0070$. The return transmission factor is also lower for AI firms, with $B_A^* = 1.6518$ compared with $B_N^* = 1.8178$. Therefore, the estimates do not imply that AI firms are harder to learn about in every dimension. Rather, they suggest that observed AI cash flow signals are noisier and receive less weight in belief updating. Whether this generates a positive AI premium gap also depends on the realized belief errors and expected cash flow shocks, which I examine in the next section with Carter–Kohn simulation.

Panel C shows that the model closely matches the variances of the log price-dividend ratios. For AI firms, the empirical and model-implied variances are 0.9813 and 0.9811, respectively. For non-AI firms, they are 1.0560 and 1.0561. The model also reproduces the higher mean squared forecast error for AI firms. The model-implied values are 0.1689 for AI firms and 0.1359 for non-AI firms, compared with empirical values of 0.1805 and 0.1223. However, the model does not do well in reproducing the empirical ordering of the expected return gap variances. In the data, the variance is 0.0285 for AI firms and 0.0151 for non-AI firms, while the model-implied values are 0.0165 and 0.0230, respectively.

4.4. *Carter-Kohn Simulation*

4.4.1. **Simulation steps**

With the estimated parameters, I conduct a simulation to quantify Channel 2 using the Carter-Kohn backward smoothing method following [Carter and Kohn \(1994\)](#). The intuition of this method is as follows. The Kalman filter forms beliefs in real time from 2009 to 2024 using dividend growth signals as they arrive, and once the full series of filtered posteriors is in hand, I draw the latent states backward from these posteriors, each draw conditional on the draw for the following year. I use this simulation method to test whether observed cash-flow dynamics are consistent with sufficiently positive expected cash-flow shocks that can explain the positive AI premium gap.

First, I construct the signal using actual annual portfolio dividend growth data for the

AI portfolio (fifth quintile) and the non-AI portfolio (first quintile):

$$s_{i,t} \equiv \Delta d_{i,t} - \mu_d. \quad (59)$$

Under the model timing, $s_{i,t}$ provides information about $\eta_{i,t-1}$. For each of 10,000 simulations, I randomly draw the signal path for each year from 2009 to 2024 using the empirical distribution of individual firms within each portfolio.

Second, I run the Kalman filter using the signal path to recover the date- t belief $\hat{\eta}_{i,t}$ and conditional variance $P_{i,t}$. The filter first uses $s_{i,t}$ to update $\eta_{i,t-1}$ and then predicts $\eta_{i,t}$.

Third, I compute the steady-state return transmission coefficient

$$B_i^* = 1 + \kappa_1 \frac{\rho_i}{1 - \kappa_1 \rho_i} G_i^*, \quad (60)$$

using the estimated persistence and steady-state Kalman gain.

Fourth, I simulate ex post expected latent states using the Carter-Kohn backward smoothing method. For each simulation, I draw 50 latent state paths:

$$\eta_{i,0:T}^{(j)} \sim p(\eta_{i,0:T} \mid s_{i,1:T}), \quad j = 1, \dots, 50. \quad (61)$$

Because the last observed signal $s_{i,T}$ measures $\eta_{i,T-1}$, the terminal state $\eta_{i,T}$ is a one-step-ahead draw conditional on the available signals. This yields 500,000 conditional state path draws over the 2009 to 2024 sample. Fifth, I map the state draws into the expected cash flow shock gap (or epsilon gap) and the AI premium gap:

$$\epsilon_{A,t} - \epsilon_{N,t} \equiv (\eta_{A,t}^{(j)} - \rho_A \eta_{A,t-1}^{(j)}) - (\eta_{N,t}^{(j)} - \rho_N \eta_{N,t-1}^{(j)}), \quad (62)$$

$$\Pi_t^{gap,(j)} = B_A^*(\eta_{A,t}^{(j)} - \hat{\eta}_{A,t}) - B_N^*(\eta_{N,t}^{(j)} - \hat{\eta}_{N,t}), \quad t = 1, \dots, T. \quad (63)$$

Last, I aggregate outcomes within each simulation that contains 16 years:

$$\bar{\Pi}^{gap} = \frac{1}{TJ} \sum_{t=1}^T \sum_{j=1}^J \Pi_t^{gap,(j)}. \quad (64)$$

The 10,000 simulations therefore produce a model implied distribution for the AI premium gap, together with ex post expected cash flow shock gap (or epsilon gap).

4.4.2. Simulation results

Figure 3 presents the simulation results. I define the ex post expected cash flow shock gap (or the epsilon gap) as

$$\epsilon_{A,t} - \epsilon_{N,t} = (\eta_{A,t} - \rho_A \eta_{A,t-1}) - (\eta_{N,t} - \rho_N \eta_{N,t-1}), \quad (65)$$

and find a positive relationship between the mean of epsilon gap and the cross-sectional AI premium gap. Each point in the simulation graph represents a 16-year average, matching the 2009 to 2024 empirical sample period. A mean epsilon gap of 0.004 is nearly sufficient to generate a positive mean AI premium gap. The length of the sample is central to reading this figure. Unconditionally the belief error has mean zero, so in a long enough sample the average epsilon gap and the average AI premium gap are both close to zero with high probability. Over 16 years they are not. The simulation delivers the sampling distribution of 16-year averages at the estimated parameters, and an average gap of the magnitude observed in the data sits well inside that distribution.

4.4.3. Comparison with actual AI premium gap

Since the epsilon gap as described above is an ex-post object estimated from backward simulation using real-world dividend data, it is natural to ask whether the model-implied epsilon gap can match the actual AI premium gap. To study this question, ideally, I need to plot the actual AI premium shock gap against the model-implied epsilon gap over time,

and see if there is any positive comovement over years.

The epsilon gap is a shock, so its empirical counterpart has to be a shock as well. The level of the AI premium gap is not the right object, because it moves with long-run means and with persistence that has nothing to do with news arriving in a given year. I therefore remove persistence from the actual gap series and from the model-implied gap series separately, and normalize both by their volatility so the two are on a comparable scale.

To construct the empirical analogue of the actual AI premium shock gap, I sort firms each month into AI Exposure quintiles and form value-weighted portfolios of the highest and lowest quintiles. For each portfolio I compute both econometric-based objective excess returns and analyst-based subjective excess returns, using the same long-term and short-term measures as in the AI premium gap estimation. A firm-month enters only if both the objective and the subjective measure are non-missing, so the two portfolios are built from the same firms with the same weights. The difference in objective returns between the high- and low-exposure portfolios gives the objective AI premium, and the analogous difference in subjective returns gives the subjective AI premium; their difference is the monthly actual AI premium gap, which I aggregate to annual frequency by simple yearly averages to match the annual frequency of the model-implied epsilon gap.

To isolate the unexpected component, I fit a single AR(1) with an intercept to the annual AI premium gap itself, and I standardize the residual innovation by the residual standard deviation of that regression. The actual AI premium shock gap is this standardized innovation. Fitting one AR(1) to the gap keeps the shock on a single scale and removes the level and the persistence of the object I care about. Constructed in this way, the actual AI premium shock gap captures whether the AI premium gap in a given year is larger or smaller than its own level and persistence would predict. I then compare this series with the model-implied epsilon gap, which is purged of AR(1) persistence and standardized analogously, so that the comparison focuses on matching innovations rather than raw levels that may differ mechanically. Given the short annual sample and the noise inherent in year-

to-year innovations, I plot 5-year moving averages of both series to focus on their smoother medium-run comovement; because the AR(1) specification requires a lag, the first year of the sample (2009) is omitted. In [Internet Appendix IA10](#), I provide results for alternative moving average horizons, statistical tests of the comovement between the two series, and full details of each construction step.

[Figure 4](#) presents the results. The correlation between the two series using short-term spreads (machine-learning-implied expected returns minus analyst target-price-implied returns) in the top panel is 0.664 and the long-term spreads (objective ICC minus subjective ICC) in the bottom panel is 0.718. The actual AI premium shock gap shows strong positive comovement with the model-implied shock gap, providing supportive evidence for Channel 2 in the model, that is, there have been more positive ex post expected cash flow shocks for high AI Exposure firms. The cash flow shock gap, in both the actual and model-implied series, becomes noticeably more positive in 2022–2024 relative to earlier years. This lines up well with the launch of ChatGPT at the end of 2022, which can be interpreted as a positive shock to firms with higher AI Exposure ([Eisfeldt et al., 2023](#)). Since analysts do not fully update due to uncertainty, objective expected returns are higher than subjective expected returns.

5. Implications

As of now, I have shown that a Bayesian learning model with two channels can help explain the variations in the *AI premium gap*. This section applies the Bayesian learning model to two other contexts, the value premium and the dot-com era of the 1990s, for external validity, and discusses the implications for the bubble debate. [Section 5.1](#) studies whether the model can help explain the variation of the *value premium gap*, defined in a similar way to the AI premium gap. [Section 5.2](#) then turns to the dot-com era. The dot-com era is a useful second test for a specific reason. It is an episode in which the bubble question was

debated intensely at the time and is still debated today, but the circumstances differ from the AI era in the technology involved, in the firms involved, and in the fact that the outcome is known ex post. The question is how the model fares during the dot-com period. I identify the firms affected by the dot-com boom as IT-exposed firms, and I define the *IT premium gap* for those firms in a similar way as the AI premium gap. The two settings deliver different verdicts. The model explains the value premium gap but fails on the IT premium gap in the dot-com era, and I argue that the failure offers useful insight. Section 5.3 discusses the implications of this paper for the bubble debate in the context of dot-com and AI.

5.1. Value Premium

First, I study whether the Bayesian learning model can shed light on the value premium. The value premium can be thought of as the excess return of a portfolio long high book-to-market stocks and short low book-to-market stocks (see, e.g., [Fama and French, 1993](#); [Zhang, 2005](#); [Hou, Xue, and Zhang, 2015](#); [Fama and French, 2021](#); [Gonçalves and Leonard, 2023](#); [Zhang, 2024](#)). I re-estimate the model by comparing value and growth firms. I define value firms as those in the highest book-to-market quintile and growth firms as those in the lowest quintile.

[Figure 5](#) presents the estimated results. The model also fits the data well in this context, with a correlation of 0.638 between the model-implied series and the actual-data generated series. What seems interesting is that value firms experience more negative cash flow shocks in the second half of the sample period, especially in the 2010s and 2020s.

[Fama and French \(2021\)](#) documents that the value premium has declined in recent decades. The existing literature offers several possible explanations for this decline. For example, the rise of intangible assets has not been fully captured when book-to-market is measured using traditional accounting data (see, e.g., [Kahle and Stulz, 2017](#); [Peters and Taylor, 2017](#); [Eisfeldt, Kim, and Papanikolaou, 2022](#); [Crouzet and Eberly, 2023](#); [Fahlenbrach, Sanz, and Stulz, 2026](#)). The listing gap literature highlights that the decline in the

number of publicly listed firms since 1996 can affect the composition of value and growth firms (see, e.g., [Gao, Ritter, and Zhu, 2013](#); [Doidge, Karolyi, and Stulz, 2017](#); [Binfàè, Brown, and Polk, 2020](#); [Doidge et al., 2025](#)). [Gonçalves and Leonard \(2023\)](#) further points out that book equity is no longer a good proxy for cash flow generation.

In my paper, the simulation results provide another potential explanation. Even when cash-flow uncertainty is similar across value and growth firms, value firms have experienced more positive cash flow shocks in the early part of the sample, but more negative cash flow shocks in the later period. This pattern directly helps explain why the value premium has weakened over time.

This points to something more general. Bayesian learning has clean asymptotic properties. Beliefs converge, and the belief error averages to zero. Those properties are often invoked to argue that a wedge between objective and subjective expectations cannot persist. The samples available in asset pricing are rarely long enough for those properties to become apparent. The AI sample studied here is 16 years, and the dot-com run-up is shorter still. Over horizons of this length the average belief error is a random variable with real dispersion rather than a number close to zero, so a persistent wedge is what a fully rational Bayesian benchmark predicts rather than evidence against it. [Section 4.4.2](#) makes the point quantitative in the AI setting. The value premium makes it concrete. A run of cash flow shocks that would wash out asymptotically is long enough here to move the premium for two decades.

5.2. *Dot-Com Era*

Next, I test how the model fares in the dot-com era. The exercise asks the same question as in the AI setting, in a period where the bubble debate was at least as intense and where the circumstances were different. I identify the firms affected by the dot-com boom as IT-exposed firms. Following [Jorgenson, Ho, and Stiroh \(2005\)](#), I classify a firm as belonging to the IT sector based on its four-digit Standard Industrial Classification (SIC) code. Specifically, a firm is defined as belonging to the IT sector if its four-digit SIC code falls within one

of the following ranges: 3570 to 3579 (computer and office equipment), 3660 to 3669 (communications equipment), 3670 to 3679 (electronic components and accessories), or 7370 to 7379 (computer programming, data processing, and other computer-related services). These are the firms whose valuations sat at the center of the dot-com run-up. I re-estimate the Bayesian model with two groups of firms, IT sector and non-IT sector, and I refer to the resulting object as the *IT premium gap*, the dot-com counterpart of the AI premium gap.

IT is the general-purpose technology here, and the dot-com run-up is the valuation episode within it that the bubble discussion is about. The literature typically identifies 1971 as the beginning of the IT era (see, for example, [Perez, 2002](#)). However, there are some limits to data availability. The econometric-based objective ICC measure and the machine-learning implied expected returns both start in 1972. Analyst based ICC starts in 1982, consistent with the coverage of analyst earnings forecasts in IBES. Analyst one-year-ahead target-price-implied expected returns with good coverage are available only from 1999 onward, so for the following analysis, I focus on long-term spreads (objective ICC as in [Gonçalves, 2021](#) minus analyst ICC as in [Eskildsen et al., 2026](#)). I report the longer sample for context, but the period of interest is the dot-com run-up of the 1990s. That is where the bubble discussion was concentrated, and that is where the model is tested.

I re-estimate the dynamic model for IT and non-IT firms and plot the actual IT premium shock gap against the model-implied cash flow shock gap, following the same approach as in the AI context (see, [Figure 4](#)). [Figure 6](#) shows the estimated results. The top panel covers the full sample, and the bottom panel zooms in on the dot-com run-up from 1989 to 1999, which is the period of interest. Unlike in the AI setting, the model does not explain the variation in the IT premium gap well. The two series do not track each other over the full sample, and they move almost entirely in opposite directions from 1994 to 1999, during the later part of the dot-com run-up. The correlations between the actual IT premium shock gap and model-implied cash flow shock gap for the full sample and the 1989–1999 subperiod are 0.110 and -0.685, respectively. The disagreement is not a matter of degree. Within the

model, the shock to the objective-minus-subjective spread is driven by cash flow news, so the two series can only comove positively. At the estimated parameters, the model-implied correlation is approximately 0.88, and it remains positive for any admissible parameter values because the gap loads on the cash flow innovations with positive coefficients $B_i^* > 0$ and the measurement noise is independent of those innovations. A negative correlation in the data is therefore a rejection of the cash flow learning mechanism in this sample, rather than an alternative calibration of it.

The timing of the disagreement is telling. From 1994 to 1999, the model-implied cash flow shock gap rises from approximately -0.45 to 0.47, while the actual IT premium shock gap falls from approximately zero to -0.37 and reaches its lowest point of -0.67 in 1998. From 1994 through 1998, the model-implied series increases every year while the actual series decreases every year. Thus, during the period closest to the peak of the dot-com run-up, the model reads relative cash flow news as improving while the actual IT premium shock gap becomes increasingly negative. Prices moved in a way that past cash flows cannot justify within the model.

Two clarifications are worth stating. First, the model does reproduce the negative level of the IT premium gap. The model-implied IT premium gap is negative in nearly every year of the sample, matching the significantly negative *IT producer premium gap* I estimate in Section 5.3. What the model cannot reproduce is the year-to-year variation, and the dot-com era is where it fails most clearly. Second, the failure provides useful evidence. The same mechanism that accounts for the variation in the AI premium gap does not account for the dot-com era, which indicates that the variation in the IT premium gap was not mainly driven by cash flow news. This is consistent with an ex post dot-com bubble (see, e.g., Ofek and Richardson, 2003; Pástor and Veronesi, 2006; Griffin et al., 2011), which I test more formally and discuss in more detail in Section 5.3.

5.3. Bubble Discussion

The main finding of a positive AI premium gap has direct implications for the AI bubble interpretation. In the Gordon Growth Model identity discussed earlier, equating prices under objective and subjective beliefs gives $\mathbb{E}[r] - \widehat{\mathbb{E}}[r] = \mathbb{E}[g] - \widehat{\mathbb{E}}[g]$. The left-hand side is the objective-minus-subjective expected returns spread, while the right-hand side is the objective-minus-subjective expected growth spread. A bubble-like interpretation is naturally associated with subjective growth expectations being too high relative to objective expectations, so that $\widehat{\mathbb{E}}[g] > \mathbb{E}[g]$. This implies $\widehat{\mathbb{E}}[r] > \mathbb{E}[r]$, or equivalently a negative objective-minus-subjective expected returns spread, or a negative AI premium gap. In contrast, the positive AI premium gap documented in this paper points in the opposite direction. Econometric-based objective expected returns are higher than the subjective beliefs, suggesting that subjective growth expectations for AI firms sit below the objective benchmark rather than above it, which is inconsistent with a simple AI bubble story driven by overoptimism on growth.

A natural question is whether the empirical framework we have seen in Section 2, which directly measures the objective-subjective expected returns spread, produces a different pattern in an episode widely viewed ex post as the dot-com bubble. I therefore apply the same empirical design to the dot-com era from 1990 to 1999, before the price crash, and examine what expected returns spread appears for the dot-com era.

For robustness, I examine two cases. Both are ways of identifying the firms affected by the dot-com boom, the first by what a firm produces and the second by what a firm adopts. The first compares IT producers and non-IT firms, where IT producers are defined as firms operating in the IT sector following [Jorgenson, Ho, and Stiroh \(2005\)](#). Using a similar panel regression method as in the AI setting, I find a negative and statistically significant *IT producer premium gap*. The second examines broader IT capital investment by studying IT adopters while excluding IT producers, following [Chun et al. \(2008\)](#). I find no evidence of a significant *IT adopter premium gap* among IT adopters. These results suggest that the

AI era differs from the dot-com era in the relation between the econometric-based objective and analyst-based subjective expected returns. If anything, even if the dot-com era can be interpreted as a bubble ex post, the evidence in this paper does not support a comparable bubble interpretation for the AI era.

5.3.1. IT producer premium gap

First, I focus on comparing IT producers in the IT sector defined in [Jorgenson, Ho, and Stiroh \(2005\)](#) to those in the non-IT sectors, over the dot-com window from 1990 to 1999. Overall, I find a significantly negative IT premium gap after adding controls and industry fixed effects. Following [Jorgenson, Ho, and Stiroh \(2005\)](#), I classify a firm as belonging to the IT sector based on its four-digit Standard Industrial Classification (SIC) code same as Section 5.2. Because analyst target-price data are limited in the early sample, I focus on the long-term spread (objective ICC following [Gonçalves, 2021](#) minus analyst ICC as in [Eskildsen et al., 2026](#)) in estimating the IT premium gap.

[Figure 7](#) Panel (a) presents the IT producer premium gap estimation results. The spread between objective and subjective expected returns is modeled as a linear function of firm-level *isITProducer* and standardized firm characteristics:

$$Spread_{i,t} \equiv \mathbb{E}_t[r_{i,t+1}] - \widehat{\mathbb{E}}_t[r_{i,t+1}] = \text{FEs} + \beta \times isITProducer_i + \text{Controls} + \varepsilon_{i,t}, \quad (66)$$

where β captures the IT producer premium gap and *isITProducer_i* denotes firm level dummy variable that indicates whether a firm belongs to the IT sector. The interpretation differs from the baseline specification for AI premium gap. Here, β measures the ex-ante expected return spread differential between IT and non-IT firms. The IT producer premium gap is significantly negative after adding controls and industry fixed effects.²³ The same pattern

²³I have also estimated the IT producer premium gap over a shorter window, 1995 to 1999. The estimates are negative throughout and close to the baseline magnitudes. With controls and time fixed effects they are significant at the 1% level, and adding industry fixed effects leaves the point estimates nearly unchanged with standard errors roughly $\sqrt{2}$ larger, reflecting the halved number of monthly cross sections.

holds for the spread of z-scores in [Figure IA5.11](#).

Next, I also estimate the forecast error gap in the context of the two IT cases, in a similar way as in [Section 3](#) for AI era. [Table 4](#) column (1) presents the difference between IT and non-IT analyst forecast bias. For each forecast horizon h , forecast bias difference between IT and non-IT sector is estimated using the following regression:

$$e_{i,t+h|t} \equiv y_{i,t+h} - \mathbb{F}_t y_{i,t+h} = a_h + \beta_{(1)} \times isITProducer_i + u_{i,t+h}, \quad (67)$$

where $y_{i,t+h}$ denotes realized earnings per share for firm i at horizon h , and $\mathbb{F}_t y_{i,t+h}$ is the analyst forecast issued at time t . Although, in longer horizons, $\beta_{(1)}$ is more negative, it shows that there is no evidence that IT sector exhibits significantly lower analyst forecast bias than non-IT sectors.

Overall, in the context of the dot-com era when comparing IT producers in the IT sector with non-IT firms, I find evidence of a significantly negative IT producer premium gap, but not significantly lower analyst forecast bias for IT producers than non-IT firms.

One concern is that this approach captures firms producing the technology itself but does not necessarily reflect the broader exposure of firms that adopt the technology in their operations. In contrast, the *AI Exposure* measure used in previous sections captures both producers and adopters through firm level discussions of AI use or investment. Ideally, one would construct a comparable firm level measure of IT exposure, but conference call transcript coverage is limited in the early years of the dot-com era, making such an approach less feasible. To address this limitation and show robustness, the next section examines IT adoption outside the IT producing sector using industry level measures of IT intensity.

5.3.2. IT adopter premium gap

I follow [Chun et al. \(2008\)](#) to study whether there is an IT premium gap during the dot-com era for traditional industries (excluding IT producer) associated with higher IT investment

intensity. This captures firms outside the IT sector that were affected by the dot-com boom through their own IT investment. The main data source for industry-level IT investment is the Bureau of Economic Analysis (BEA) Nonresidential Detailed Estimates data that track 96 asset classes from 1972 to 1999 by two-digit SIC industry. The net stocks are estimated using a perpetual inventory model as [Hall \(1990\)](#). By definition, industry j 's stock of asset k at time t is

$$K_{j,k,t} = (1 - \delta_k)K_{j,k,t-1} + I_{j,k,t}, \quad (68)$$

where δ_k is an asset-specific depreciation rate and $I_{j,k,t}$ is the industry's investment on type k assets in year t .

As in [Chun et al. \(2008\)](#), I define IT capital as seven classes of computer hardware (mainframe computers, personal computers, direct access storage devices, computer printers, computer terminals, computer tape drives, and computer storage devices) and three classes of software (pre-packaged software, custom software, and own-account software).

Industry j 's IT intensity in year t is then defined as the stock of IT capital relative to the stock of non-IT capital:

$$ITIntensity_{j,t} \equiv \frac{\sum_{k \in IT} K_{j,k,t}}{\sum_{k \notin IT} K_{j,k,t}}. \quad (69)$$

Figure 7 Panel (b) presents the IT adopter premium gap estimation results, with the above industry-level measure of IT Intensity. Specifically, the IT adopter premium gap is estimated in the following setup:

$$Spread_{i,t} \equiv \mathbb{E}_t[r_{i,t+1}] - \widehat{\mathbb{E}}_t[r_{i,t+1}] = \text{FEs} + \beta \times ITIntensity_{j,t-1} + \text{Controls} + \varepsilon_{i,t}, \quad (70)$$

where β captures the IT adopter premium gap and $ITIntensity_{j,t-1}$ denotes lagged, standardized industry level *IT Intensity*. The estimation results show that there is no significant IT adopter premium gap.

Table 4 column (2) presents the differential analyst forecast bias between high and low IT

Intensity adopters. For each forecast horizon h , it is estimated using the following regression:

$$e_{i,t+h|t} = a_h + \beta_{(2)} ITIntensity_{j,t} + u_{i,t+h}, \quad (71)$$

where $e_{i,t+h|t}$ denotes the forecast error for firm i at horizon h , and $ITIntensity_{j,t}$ denotes the standardized IT investment intensity of industry j at time t . The estimation results show that analyst forecast errors do not vary significantly with IT Intensity.

Overall, in the context of the dot-com era when focusing on IT adopter firms outside the IT sector, I do not find evidence of a significant IT adopter premium gap, nor significantly different analyst forecast bias across different levels of IT Intensity among these IT adopters.

6. Conclusion

We observe high valuations among AI stocks, but whether these valuations indicate an AI bubble remains unclear ex ante. To shed light on this question, this paper introduces a new empirical framework to directly measure the wedge between econometric-based objective expected returns and analyst-based subjective expected returns for AI firms relative to non-AI firms, which I call the *AI premium gap*.

The logic of the framework follows the bubble and subjective belief literature. Bubble-like pricing is often associated with high prices supported by overly optimistic beliefs. When prices are high relative to fundamentals, objective expected returns are low. At the same time, investors who are overly optimistic about future cash-flow growth may still view those prices as justified and perceive expected returns to be high. A negative gap, in which subjective expected returns exceed objective expected returns, is therefore consistent with bubble-like pricing. A positive gap is not. I find the opposite of the bubble prediction in the data. The AI premium gap is robustly positive.

A simple accounting identity turns this return-based finding into a testable prediction about analyst forecasts. Since prices and dividends are observed, the Gordon Growth Model

valuation can be written under objective expectations and under coherent subjective expectations as $P = D/(\mathbb{E}[r] - \mathbb{E}[g]) = D/(\hat{\mathbb{E}}[r] - \hat{\mathbb{E}}[g])$, where hats denote subjective measures, which gives $\mathbb{E}[r] - \hat{\mathbb{E}}[r] = \mathbb{E}[g] - \hat{\mathbb{E}}[g]$. Read through this lens, a positive AI premium gap suggests that subjective expected growth is below objective expected growth for AI firms relative to non-AI firms. If so, realized earnings should exceed analyst forecasts by more for AI firms relative to non-AI firms. I confirm this prediction in the data. The *forecast error gap* is positive, which provides direct, model-free evidence consistent with the positive AI premium gap.

I then study why this positive gap arises. Building a Bayesian learning model, I show that two channels jointly explain the gap. The first is cash-flow uncertainty. When firms are more exposed to AI, their future cash flows are harder to evaluate, so analysts receive noisier signals and update more slowly. The second is positive expected cash flow shocks. AI firms have experienced more positive cash flow shocks in recent years, especially in the early 2020s. Because analysts learn gradually from past outcomes, these positive shocks make objective expectations move ahead of subjective expectations. Together, higher cash-flow uncertainty and positive expected cash flow shocks jointly generate a positive AI premium gap.

The model’s performance in other settings also highlights some important implications. When applied to the value premium context, the model matches the observed dynamics in the data well. Relative cash-flow shocks in the long run were negative for value firms relative to growth firms during the 2010s and 2020s. This pattern offers a new angle to understand the value premium decline in recent decades, documented in [Fama and French \(2021\)](#). In the IT setting, by contrast, the model fails. The model can only generate a positive comovement between the shock to the objective-minus-subjective spread and cash flow news, while the comovement in the data is strongly negative, especially in the late 1990s. The failure indicates that the variation in the IT premium gap during the dot-com era was not driven by cash flow news.

The dot-com application also connects these model-implied cash-flow dynamics to the

IT premium gap evidence. Using the same empirical framework as in the AI analysis, I find a negative *IT premium gap* comparing IT producers and non-IT firms. Together with the inability of the learning model to account for its variation, the negative IT premium gap provides ex post evidence consistent with a dot-com bubble. By contrast, the positive *AI premium gap* points in the opposite direction. The evidence through 2024 therefore suggests a fundamental difference between the two eras.

Overall, the findings speak directly to the AI bubble debate. The evidence is difficult to reconcile with a simple bubble story in which high AI valuations are supported by overly optimistic beliefs. If anything, the positive AI premium gap points in the opposite direction. The objective expected return is higher than the subjective expected return for AI firms relative to non-AI firms. I stress that a wedge between objective and subjective expected returns does not by itself establish or rule out a bubble. What the framework delivers is a transparent, sign-based diagnostic for whether the pricing pattern in a given episode is consistent with a bubble story.

Several limitations point to future research. First, my *AI Exposure* measure is based on conference call transcripts, which is useful for capturing investor beliefs, as discussed in [Babina \(2026\)](#), but I cannot fully test robustness across all possible exposure measures. Second, the model shows that individual rationality, accounting for limited information, alone can explain the AI premium gap through cash-flow uncertainty and expected cash flow shocks, though behavioral mechanisms are not ruled out. Third, the sample ends in 2024, and limited IBES coverage makes it difficult to extend analyst-based expected return measures to earlier periods.

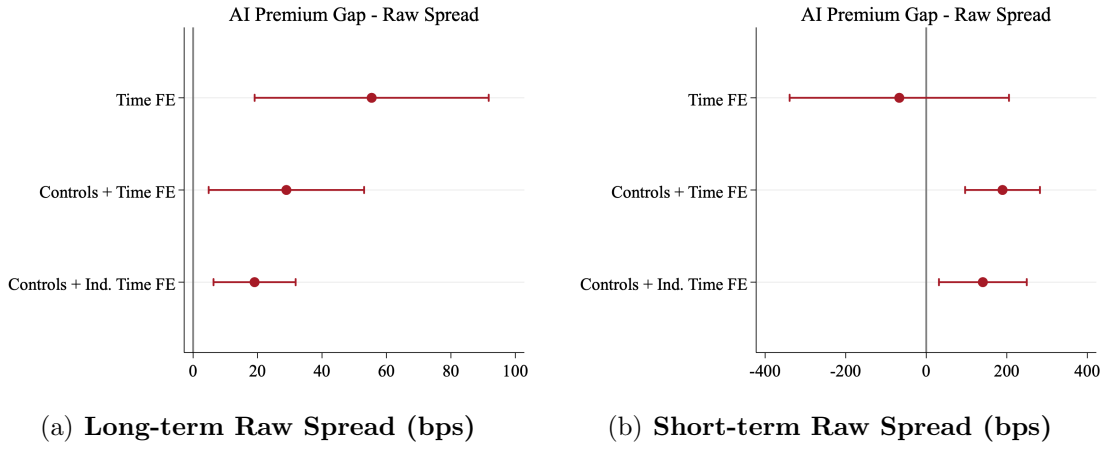
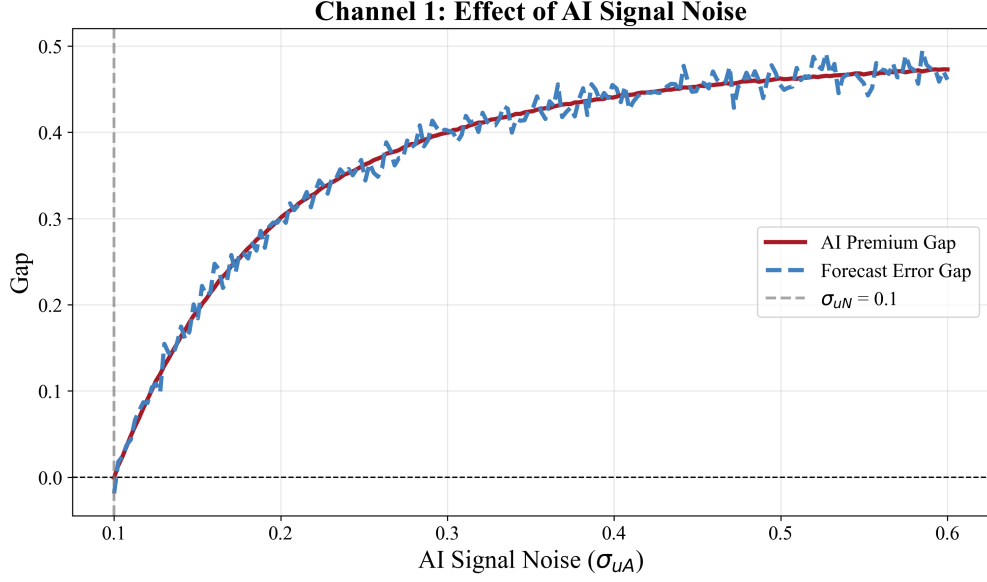
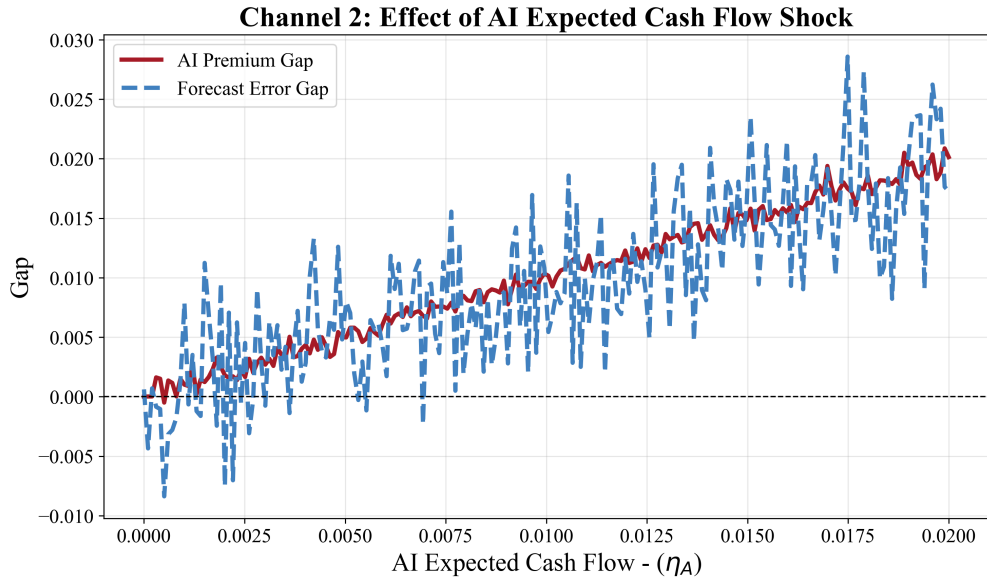


Figure 1. AI Premium Gap. The figure reports the estimated annual *AI premium gap* from 2009 to 2024, based on panel regressions of the spread between objective expected returns and subjective expected returns on lagged, standardized *AI Exposure*, following a similar approach to [Eskildsen et al. \(2026\)](#). Each panel reports three specifications: Time fixed effects (FE) only, Time FE with controls, and Time FE with controls and Industry FE. Controls include size, BE/ME, ROE, R&D, age, and asset growth. Standard errors are clustered by firm, month, and Fama–French 49 industry. Panel (a) uses the long-term spread, defined as econometric-based objective ICC following [Gonçalves \(2021\)](#) minus analyst-based subjective ICC following [Eskildsen et al. \(2026\)](#). Panel (b) uses the short-term spread, defined as machine-learning-implied expected returns following [Bastianello \(2022\)](#) minus analyst one-year-ahead target-price-implied returns following [Loudis \(2024\)](#).



(a) Channel 1



(b) Channel 2

Figure 2. Two-Period Illustrative Model. The figure summarizes two channels in the two-period illustrative model that generate a positive *AI premium gap*, defined as the difference between objective and subjective expected returns for AI firms relative to non-AI firms. *Channel 1 (Noisier Cash Flow Signals)* is illustrated in a case where latent expected cash flow realizations are the same, but AI firms have higher signal noise, leading to a lower Kalman gain and slower belief updating. In this case, the *AI premium gap* reflects a larger wedge between objective and subjective expectations about AI cash flow news. *Channel 2 (Positive Latent Expected Cash Flow)* is illustrated in a case where signal precision is the same across AI and non-AI firms, but AI firms experience larger realizations of the latent expected cash flow. In this case, the *AI premium gap* arises because expected cash flow exceeds investors' posterior beliefs by more for AI firms. In the model, risk premium terms cancel between objective and subjective expectations, so the *AI premium gap* is driven entirely by differences in learning about the latent expected cash flow under uncertainty.

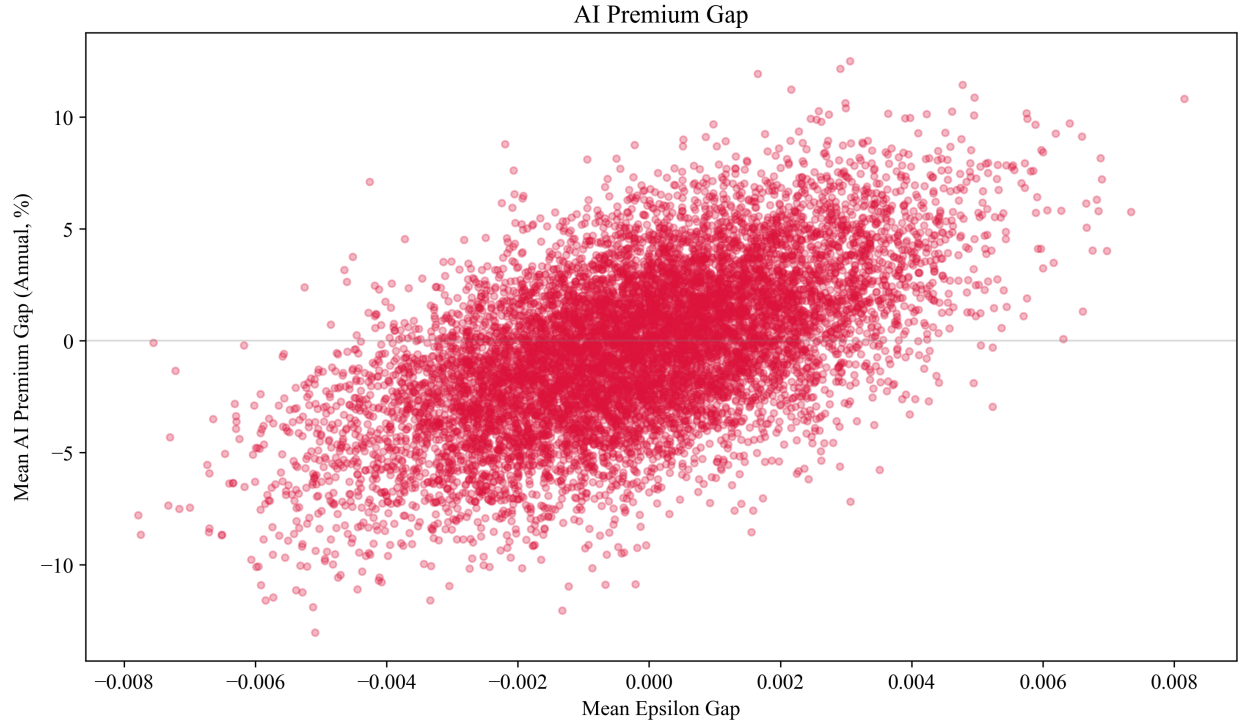


Figure 3. Dynamic Model Simulation. This figure plots the simulated relationship between the *AI premium gap* and the expected cash flow shock gap (or epsilon gap). The AI premium gap is the difference between the objective AI premium and the subjective AI premium. The expected cash flow shock gap, or epsilon gap, captures the difference in expected cash flow shocks between AI and non-AI firms and is defined as $\epsilon_{A,t} - \epsilon_{N,t} = (\eta_{A,t} - \rho_A \eta_{A,t-1}) - (\eta_{N,t} - \rho_N \eta_{N,t-1})$. Each point represents a 16-year average, matching the 2009 to 2024 empirical sample period.



Figure 4. Actual AI Premium Shock Gap vs. Model-Implied Shock Gap. This figure compares the actual AI premium shock gap with the shock gap implied by the dynamic model. Firms are sorted monthly into AI Exposure quintiles; the *AI premium gap* is the value-weighted Q5 minus Q1 spread in objective minus subjective expected returns. Monthly gaps are averaged to annual frequency, purged of AR(1) persistence in the gap itself, and standardized. Both series are plotted as 5-year moving averages. The red solid line is the actual AI premium shock gap (left axis). The blue dotted line is the model-implied shock gap (right axis), constructed by Carter–Kohn smoothing of realized portfolio dividend growth and processed in the same way. The top panel uses short-term spreads (machine-learning-implied expected returns as in Bastianello, 2022 minus analyst target-price-implied returns as in Loudis, 2024). The bottom panel uses long-term spreads (objective ICC following Gonçalves, 2021 minus analyst ICC as in Eskildsen et al., 2026).

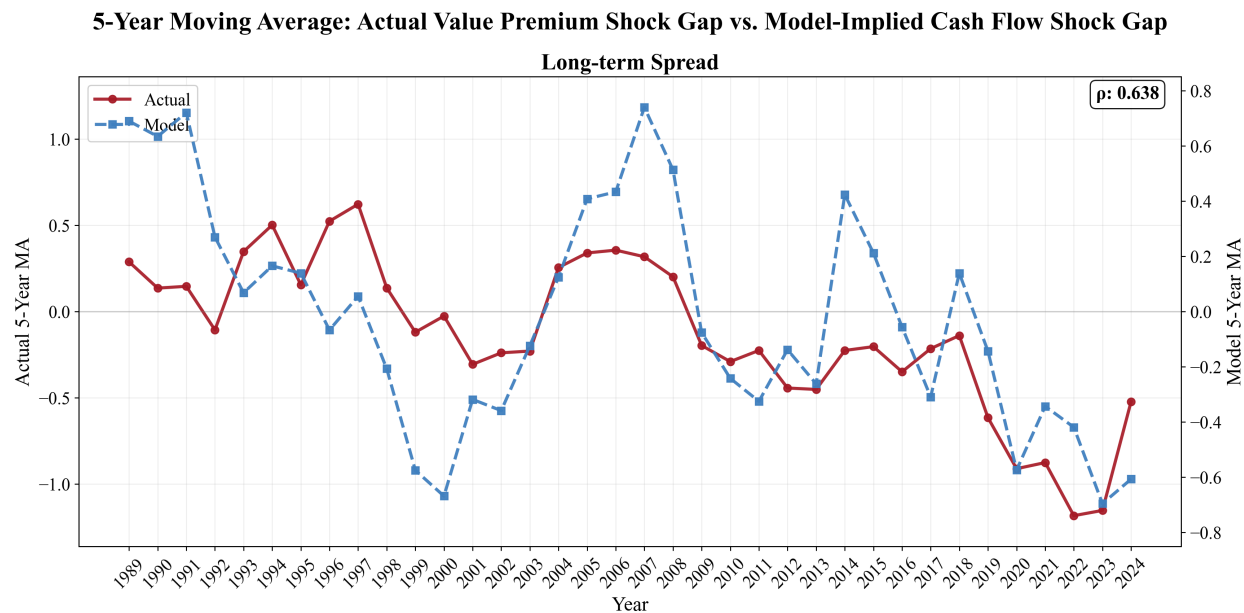


Figure 5. Actual Value Premium Shock Gap vs. Model-Implied Shock Gap. This figure compares the actual value premium shock gap with the model-implied cash flow shock gap. Firms are sorted monthly into value and growth, and the *value premium gap* is the value-weighted high-minus-low spread in objective minus subjective expected returns. Monthly gaps are aggregated to annual frequency, purged of AR(1) persistence in the gap itself, standardized, and shown as 5-year moving averages. The red solid line (left axis) is the actual gap, and the blue dotted line (right axis) is the model-implied gap. The figure uses long-term spreads (objective ICC following Gonçalves, 2021 minus analyst ICC as in Eskildsen et al., 2026); short-term spread is not available due to limited analyst target price data in early years.

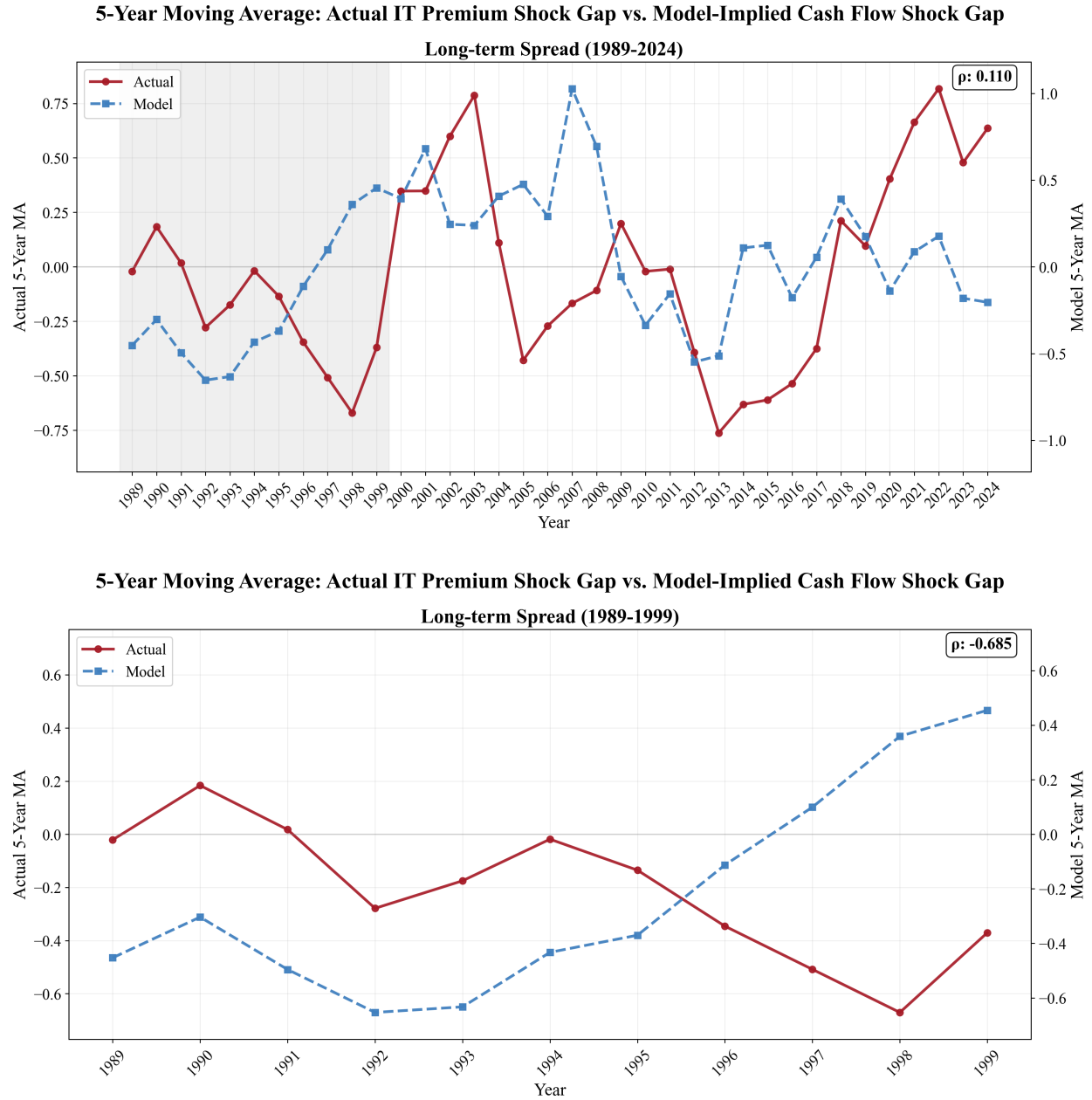


Figure 6. Actual IT Premium Shock Gap vs. Model-Implied Shock Gap. This figure compares the actual IT premium shock gap with the model-implied cash flow shock gap. Firms are sorted monthly into IT and non-IT, and the *IT premium gap* is the value-weighted IT minus non-IT spread in objective minus subjective expected returns. Monthly gaps are aggregated to annual frequency, purged of AR(1) persistence in the gap itself, standardized, and shown as 5-year moving averages. The red solid line (left axis) is the actual gap, and the blue dotted line (right axis) is the model-implied gap. The top panel covers the full sample, and the bottom panel the dot-com era from 1989 to 1999. The shaded region in the top panel marks the 1989 to 1999 dot-com era shown in the bottom panel. Both panels use long-term spreads (objective ICC following [Gonçalves, 2021](#) minus analyst ICC as in [Eskildsen et al., 2026](#)); short-term spread is not available due to limited analyst target price data in early years.

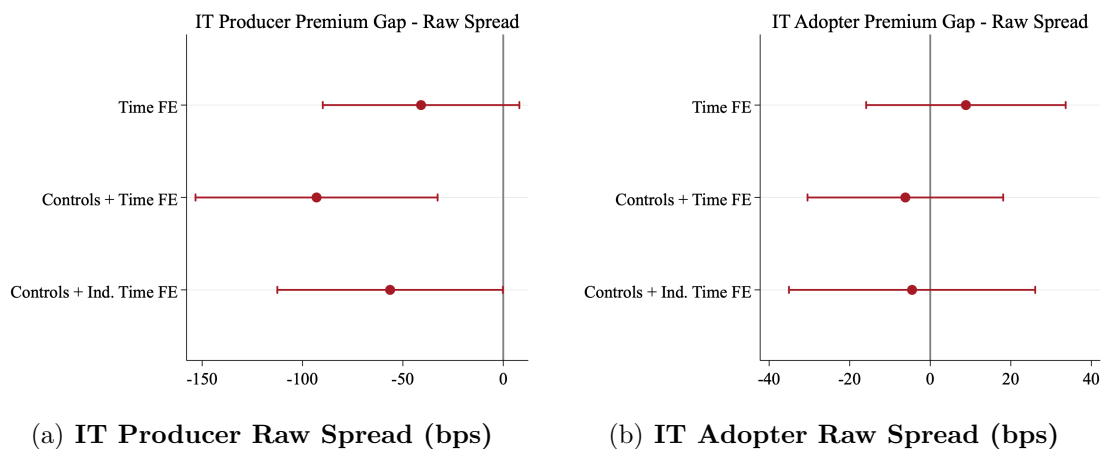


Figure 7. IT Premium Gap. The figure reports estimated annual *IT premium gaps* from 1990 to 1999, based on panel regressions of expected-return spreads on two different lagged IT measures, following [Eskildsen et al. \(2026\)](#). Panel (a) reports the *IT producer premium gap*, estimated using the IT sector dummy, while Panel (b) reports the *IT adopter premium gap*, estimated using standardized IT Intensity. Controls include size, BE/ME, ROE, R&D, age, and asset growth. Standard errors are clustered by firm, month, and Fama–French 49 industry. Both panels use raw long-term spreads, defined as objective ICC following [Gonçalves \(2021\)](#) minus analyst ICC as in [Eskildsen et al. \(2026\)](#).

Table 1
Analyst Forecast Bias: AI

This table reports analyst earnings forecast bias in cents per share across forecast horizons following [Gómez-Cram and Lawrence \(2025\)](#). Forecast bias is defined as realized earnings per share minus the analyst forecast. Rows report results by forecast horizon. Forecast bias is estimated using the regression

$$e_{i,t+h|t} \equiv y_{i,t+h} - \mathbb{F}_t y_{i,t+h} = a_h + \beta_h^{AIE} AIE_{i,t} + u_{i,t+h}, \quad (72)$$

where $y_{i,t+h}$ denotes realized earnings per share for firm i at horizon h , and $\mathbb{F}_t y_{i,t+h}$ is the analyst forecast formed at time t . The intercept a_h captures the average forecast bias at horizon h , while β_h^{AIE} captures how forecast errors vary with firm-level *AI Exposure*. $AIE_{i,t}$ is the standardized AI Exposure measure prior to forecast issuance. AI Exposure is standardized to have zero mean and unit standard deviation within each period. Standard errors are clustered by firm and calendar year-quarter. The sample period is 2009 to 2024.

	a_h (1)	β_h^{AIE} (2)
Forecast horizon: one quarter		
Coefficient	-3.17	12.65
t -statistic	[-0.67]	[2.76]
Observations	151,731	
Forecast horizon: two quarters		
Coefficient	-10.09	13.43
t -statistic	[-2.04]	[2.87]
Observations	147,419	
Forecast horizon: three quarters		
Coefficient	-17.15	16.92
t -statistic	[-3.28]	[3.47]
Observations	142,205	
Forecast horizon: four quarters		
Coefficient	-21.66	18.23
t -statistic	[-3.70]	[3.37]
Observations	137,034	

Table 2
Calibration and Target Moments for GMM Estimation

Panel A reports parameters calibrated using data from 2009 to 2024 and held fixed in the GMM estimation. Panel B reports the empirical target moments used to identify the remaining structural parameters.

Panel A: Calibrated Parameters		
Parameter	Symbol	Value
Log risk-free rate	r_f	0.0117
Market volatility	σ	0.1804
Dividend growth mean	μ_d	0.0683
AI market beta	β_A	0.9414
Non-AI market beta	β_N	0.8835
Campbell-Shiller constant	κ_0	0.0743
Campbell-Shiller discount coefficient	κ_1	0.9858

Panel B: Empirical Target Moments		
Moment	Symbol	Target
Subjective excess return (AI)	$\widehat{\mathbb{E}}_t[r_{A,t+1} - r_f]$	0.0682
Subjective excess return (Non-AI)	$\widehat{\mathbb{E}}_t[r_{N,t+1} - r_f]$	0.0737
Variance of log P/D (AI)	$\mathbb{V}(z_{A,t})$	0.9813
Variance of log P/D (Non-AI)	$\mathbb{V}(z_{N,t})$	1.0560
Variance of obj-sub spread (AI)	$\mathbb{V}(\text{Spread}_{A,t})$	0.0285
Variance of obj-sub spread (Non-AI)	$\mathbb{V}(\text{Spread}_{N,t})$	0.0151
Forecast error variance (AI)	$\widehat{\mathbb{E}}_t[FE_{A,t+1}^2]$	0.1805
Forecast error variance (Non-AI)	$\widehat{\mathbb{E}}_t[FE_{N,t+1}^2]$	0.1223

Table 3
GMM Estimation Results

Panel A reports GMM estimates of the structural parameters. Panel B reports the implied steady-state objects of the Kalman filter and the pricing coefficients for AI and non-AI firms, evaluated at the Panel A estimates. Panel C compares empirical target moments with their model-implied counterparts and reports the corresponding matching errors.

Panel A: GMM Estimated Structural Parameters			
Parameter	Symbol	Estimate	Std. Error
Risk aversion	γ	5.8006	0.9645
AI persistence	ρ_A	0.9616	0.0309
Non-AI persistence	ρ_N	0.9547	0.0182
AI innovation variance	$\sigma_{A,\epsilon}^2$	0.0007	0.0011
Non-AI innovation variance	$\sigma_{N,\epsilon}^2$	0.0009	0.0008
AI observation noise	$\sigma_{A,u}^2$	0.1340	0.0558
Non-AI observation noise	$\sigma_{N,u}^2$	0.1036	0.0354

Panel B: Implied Steady-State Objects			
Object	Symbol	AI	Non-AI
Signal noise variance	U_i	0.1628	0.1290
Conditional variance	P_i^*	0.0060	0.0070
Kalman gain	G_i^*	0.0357	0.0511
Campbell–Shiller coefficient	$A_{1,i}$	19.2433	16.9964
Transmission factor	B_i^*	1.6518	1.8178

Panel C: Moment Matching Performance			
Moment	Data	Model	Error
$\widehat{\mathbb{E}}_t[r_{A,t+1} - r_f]$	0.0682	0.0632	-0.0050
$\widehat{\mathbb{E}}_t[r_{N,t+1} - r_f]$	0.0737	0.0786	0.0049
$\mathbb{V}(z_{A,t})$	0.9813	0.9811	-0.0002
$\mathbb{V}(z_{N,t})$	1.0560	1.0561	0.0001
$\mathbb{V}(\text{Spread}_{A,t})$	0.0285	0.0165	-0.0120
$\mathbb{V}(\text{Spread}_{N,t})$	0.0151	0.0230	0.0078
$\widehat{\mathbb{E}}_t[FE_{A,t+1}^2]$	0.1805	0.1689	-0.0116
$\widehat{\mathbb{E}}_t[FE_{N,t+1}^2]$	0.1223	0.1359	0.0137

Table 4
Analyst Forecast Bias: IT

This table reports analyst earnings forecast bias (in cents per share) across forecast horizons for the period 1990-1999. Forecast bias is defined as realized earnings per share minus the analyst forecast formed at time t . For each forecast horizon h , $\beta_{(1)}$ is estimated using the following regression:

$$e_{i,t+h|t} \equiv y_{i,t+h} - \mathbb{F}_t y_{i,t+h} = a_h + \beta_{(1)} \times isITProducer_i + u_{i,t+h}, \quad (73)$$

where $y_{i,t+h}$ denotes realized earnings per share for firm i at horizon h , $\mathbb{F}_t y_{i,t+h}$ is the analyst forecast issued at time t , and $isITProducer_i$ is constructed following [Jorgenson, Ho, and Stiroh \(2005\)](#). Column (1) reports the coefficient on the *isITProducer*, which captures whether a firm is an IT producer. Standard errors are clustered by firm and calendar year-quarter. Column (2) instead uses an industry-level measure of IT adoption. In this specification, $\beta_{(2)}$ is estimated as

$$e_{i,t+h|t} = a_h + \beta_{(2)} \times ITIntensity_{j,t} + u_{i,t+h}, \quad (74)$$

where $ITIntensity_{j,t}$ denotes the standardized IT investment intensity of industry j at time t , following [Chun et al. \(2008\)](#). This variable measures IT adoption by non-IT-producing industries based on their investment in IT-related assets. Standard errors are clustered by industry and calendar year-quarter.

	IT Producer (1990–1999)	IT Adopter (1990–1999)
	(1)	(2)
Forecast horizon: one quarter		
Beta	33.45	−23.49
t -statistic	[0.66]	[−0.85]
Observations	93,669	59,739
Forecast horizon: two quarters		
Beta	−35.93	−16.11
t -statistic	[−1.18]	[−0.93]
Observations	75,722	48,177
Forecast horizon: three quarters		
Beta	−94.67	8.87
t -statistic	[−1.27]	[0.37]
Observations	67,691	42,736
Forecast horizon: four quarters		
Beta	−214.99	19.01
t -statistic	[−1.26]	[0.64]
Observations	59,931	37,481

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